Marker genes are not self-contained definitions of cell type

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Abstract

Single-cell RNA sequencing studies report genes that mark cell types, states, and conditions. These markers reflect differential expression, prior knowledge, and expert opinion and are reported as cell type–gene pairs. Atlas building often treats those pairs as transferable to global cell type definitions. We tested this assumption by building `LLMarkers`, a benchmark of 1,560 reported marker claims from seven single-cell RNA-seq studies, each linked to source evidence and supporting differential expression.[1] Reported markers were enriched among strong differential expression signals, but a fixed rank cutoff recovered only about half of them.[2] LLM extraction from manuscripts outperformed zero-shot marker generation and selection from differential expression tables, so we used it to curate 32,621 marker associations from 855 bioRxiv and Human Cell Atlas-linked papers.[3] Same-named cell types shared marker genes above background, but 99.2% of same-label cross-paper pairs did not report identical marker sets. Recurrent immune labels recovered shared marker vocabularies only in aggregate.[4] Thus, reported markers carry transferable biological signal, but they are not self-contained global definitions; their reuse requires the source context that produced them. We formalize the definition of a marker gene and prove that under this definition it is not self-contained.[5] Therefore, to facilitate the reuse of markers, we release `LLMarkersDB` as a searchable resource that preserves marker claims, mapped genes, source text, and study context.[6]

1 Introduction

Cell type annotation in single-cell RNA-seq starts with a local comparison. Cells are partitioned into clusters, then tested for differential expression against other cells in the same experiment, often with one-versus-all differential expression. Experts review the top differential expression results and use prior knowledge, the literature, reference atlases, and sometimes follow-up experiments to assign cell type labels [Macosko et al., 2015, Stuart et al., 2019, Wolf et al., 2018, Heumos et al., 2023, Abdelaal et al., 2019, Pullin and McCarthy, 2024]. This procedure produces locally defined markers where each marker is selected relative to the cells, samples, assays, and comparisons in the experiment that produced it.

Once identified, marker genes are often reported as flat cell type–gene pairs because this format is practical. Single-cell count matrices and differential expression (DE) tables are large, while a cell type–gene pair is a compact result that can be read by people and machines, moved across papers, and loaded into databases such as CellMarker [Hu et al., 2022], PanglaoDB [Franzén et al., 2019], and Cell Taxonomy [Jiang et al., 2022], or curated with tools such as MarkerGeneBERT [Cheng et al., 2024a]. Prior benchmarking has also found that public marker databases can list different markers for the same cell population [Abdelaal et al., 2019]. Even though cell ontologies provide controlled terms and identifiers for annotating cell type labels [Osumi-Sutherland et al., 2021, Tan et al., 2026], authors usually choose the reported label and its level of specificity. The label then begins to carry marker genes implicitly. For example, "brown adipocyte" evokes *UCP1*-mediated thermogenesis and "regulatory T cell" evokes *FOXP3* and *IL2RA*. This format is ultimately useful, but hides which cells were compared, how the data were processed, and why an author selected one marker rather than another high-ranking DE gene.

Cell type annotation is shaped by these two distinct but related challenges: local-to-global transfer of markers and variable cell type nomenclature. A gene that separates a cluster from the other cells in one experiment may not separate the same cell type in another experiment [Krzak et al., 2026], yet automated annotation tools depend on transferable labels, marker programs, marker databases, or trained classifiers to assign cell type names in new datasets [Kiselev et al., 2018, Aran et al., 2019, Stuart et al., 2019, Pliner et al., 2019, Zhang et al., 2019,

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CellMarker lists *UCP1* as a "10x Chromium" marker for brown adipocytes from normal adipose tissue, citing [Zhong et al., 2020], and a "sci-RNA-seq" marker for brown adipocytes from abdominal fat, citing [Ramirez et al., 2020]. The first paper profiled bone marrow with 10x Chromium and "did not detect the expression of . . . *Ucp1*." The second profiled differentiating preadipocytes with SCRB-Seq and similarly did not detect "the brown adipocyte markers. . . *UCP1*."

Ianevski et al., 2022, Conde et al., 2022, Abdelaal et al., 2019]. The second is inconsistent naming. Authors interpret local marker programs against prior knowledge to assign cell type labels. In effect, known label-to-gene associations are used to solve the inverse problem of mapping a local gene program back to a biologically meaningful name [Pliner et al., 2019, Zhang et al., 2019, Ianevski et al., 2022, Bi et al., 2026]. This makes the reported label useful, but also unstable: cell type definitions and names have often been organized study-by-study, producing variable nomenclature and limited alignment between taxonomies [Miller et al., 2020]. As we show below, different authors can assign different names to similar marker programs, and the same name to different local marker programs.

Solving the annotation problem is central to building organism-scale cell-type resolved atlases. The Human Cell Atlas set out to "define all human cell types in terms of distinctive molecular profiles" and connect those profiles to classical cellular descriptions such as location and morphology [Human Cell Atlas Meeting Participants, 2017]. That goal requires local marker claims to support global cell identities across papers, tissues, donors, assays, and disease states. As Figure 1 illustrates, a gene that separates cell types within one study may not separate the same labels after multiple studies are pooled. The same reported label can point to different marker programs. For example, "monocyte" is reported with *LYZ*, *S100A8*, *S100A9*, and *CD14*; with *S100A8*, *S100A9*, and *CD14*; and with the interferon-response genes *ISG15*, *IFI44*, and *IFIT5* [Yang et al., 2019, Yu et al., 2024, Rigby et al., 2023]. The reverse also occurs. Different labels, including "classical", "MDSC", and "macrophage", can share the same *S100A8*, *S100A9*, and *S100A12* marker program [Conde et al., 2022, Werba et al., 2023, Pelka et al., 2021]. In both cases, the label alone is not enough. The source context is needed to decide whether the relationship reflects a shared cell type, a state, a subtype, or the local comparison used in one study in order to reuse a marker in another dataset.

In this study, we quantify the extent to which locally defined marker genes apply globally and the prevalence of the naming issue. We rebuild the link between marker gene claims and their evidence by assembling `LLMarkers`, a benchmark of 1,560 cell type–gene pairs from seven single-cell RNA-seq studies, each linked to the sentence or figure where it was reported and to the differential expression comparison from which it was drawn.[8] We show that reported markers are enriched among strong differential expression signals, but cannot be recovered by a simple rank threshold.[9] To scale marker gene comparisons across studies, we compare marker generation from cell type names, marker selection from differential expression tables, and marker extraction from manuscripts with Large Language Models (LLMs). Extraction most faithfully recovers the markers that papers report.[10] Using LLM marker extraction, we curate markers from 504 bioRxiv preprints and 457 Human Cell Atlas-linked publications.[11] The resulting corpus shows that same-named cell types share more marker genes than background, but their marker sets often remain different.[12] These results show that marker genes can discriminate cell types locally without defining them globally. This matters because atlas building expects labels to transfer across studies. If labels are expected to transfer, the marker evidence and context that produced those labels must remain searchable too. We release the benchmark, bioRxiv, and Human Cell Atlas-linked records through `LLMarkersDB`, a searchable resource for atlas building that preserves marker claims and source evidence.[13]

2 Results

Marker genes reported in scRNAseq studies are local features that partition cells [Miao et al., 2020], but atlas building requires markers that can be compared across studies. We therefore first built a benchmark dataset that links reported markers to the evidence and data comparisons that produced them. This benchmark let us test whether author-reported markers can be recovered from differential expression alone, whether LLMs can recover the same claims, and whether extracted marker claims remain consistent when compared across papers.

Manually Curating Marker Genes

`LLMarkers` contains cell type marker gene associations curated from the text and figures of seven single-cell RNA-seq studies spanning six tissues: `adipose` (two studies) [Emont et al., 2022, Hildreth et al., 2021], `bone marrow` [Zhong et al., 2020], `retina` [Gautam et al., 2021], `lung` [Adams et al., 2020], `ovary` [Wagner et al., 2020], and `testis` [Shami et al., 2020]. Across these studies, `LLMarkers` contains 2,528 markers covering 168 cell types, 641 unique genes, and 1,560 unique cell type–gene pairs.[14] Marker reporting was split across manuscript text and figures: 1,151 pairs (74%) appeared only in figures, 256 (16%) appeared only in text, and 153 (10%) appeared in both.[15] Full per-study counts are summarized in **Supplementary Table S1**.

For each marker, we recorded the reported cell type label, mapped gene ID, source sentence or figure where it was reported, and supporting differential expression comparison when available. This converts a flat cell type–gene pair into a provenance-linked marker claim. It also preserves the local comparison that gave the marker meaning.

This structure matters because reported markers are not identical to DEG table entries. The overlap between curated markers and DEG tables ranged from 18% in `bone` to 97% in `lung`. In `adipose (Emont)`, the authors describe *PDGFRA* as expressed across all six adipocyte stem and progenitor cell subpopulations, but the DEG table flags *PDGFRA* for only one.[16] Marker claims therefore combine differential expression, prior knowledge, report-

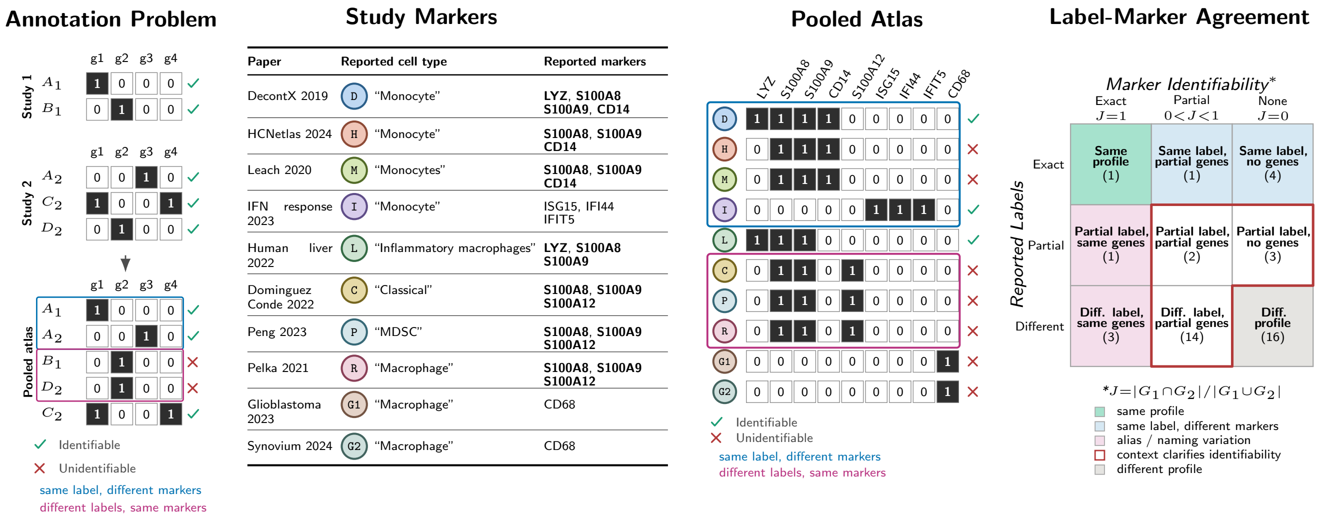

Figure 1: Cell type annotation requires distinguishing markers and labels. Left: example binary marker matrices show that marker genes selected within separate studies can become indistinguishable after studies are pooled; colored outlines mark same-label/different-marker and different-label/same-marker cases. Middle: extracted myeloid markers from a small curated corpus; bold genes are shared by at least one cell type. Right: the joint distribution compares reported label similarity with marker-gene set similarity. Parentheses count cell type pairs among the displayed examples, and marker overlap is measured by Jaccard similarity, $J = |G_1 \cap G_2|/|G_1 \cup G_2|$.

ing choices, and biological interpretation*. LLMarkers preserves those pieces so they can be evaluated directly rather than collapsed into an unsupported binary pair.

Even when a marker appears in a DEG table, its meaning depends on which comparison produced it. The same cell type–gene pair can appear in more than one table with different fold changes, and sometimes with opposite signs. Of 544 curated marker pairs that appear in at least two DEG tables within the same study, 46 (8.5%) are upregulated in one table and downregulated in another. These contradictions were concentrated in studies with multiple cell-type comparisons or metadata-stratified contrasts, including adipose (Hildreth) (39%) and lung (12%).[20] Thus, the marker claim is not just the cell type and gene. It also depends on the comparison that produced it. LLMarkers keeps that comparison linked to the reported claim.

*The authors adopted ASPC ("adipose stem and progenitor cells") as a unifying term for the Lin$^-$ stromal vascular fraction, noting that prior nomenclature—FAPs, ASCs, APCs, preadipocytes—is "complicated and inconsistent" [Emont et al., 2022]. *PDGFRA* was identified as the common marker gene for this group, a choice driven by field consensus rather than differential expression ranking. Because *PDGFRA* is broadly expressed across all six subpopulations, it does not pass the DE threshold when comparing one subpopulation against the others, and appears in the DEG table for only ASPC 1. The second adipose dataset in our benchmark [Hildreth et al., 2021] also reports *PDGFRA* as a marker for progenitor-like cells, but labels them "adipocyte precursor cell" and "preadipocyte"—entirely different names for an overlapping biology. Of the 73 combined cell type names across both studies, only three are shared.[17]

Evaluating Marker Genes

Using LLMarkers, we next asked whether reported markers can be recovered from differential expression strength alone. We matched each curated marker to its entry in the corresponding DEG table, extracted its log fold change, and ranked genes within each cell type comparison. Across all seven datasets, 303 text markers and 763 image markers could be matched to DEG entries. All were statistically significant, but their fold changes varied widely (median LFC 1.19, IQR 0.66–2.00).[21]

Reported markers were enriched near the top of DEG lists, but they were not confined to the strongest genes. The median rank was 40 for text markers and 50 for image markers. About half of reported markers fell within the top 50 DEGs (53% text, 50% image), and about two-thirds fell within the top 100 (66% text, 62% image).[22] The remaining markers came from deeper in the ranked list, sometimes beyond rank 500. Although text and figure markers were often different genes (median Jaccard 0.33), their per-cell-type fold changes were strongly correlated (Spearman $\rho = 0.80$, $p < 10^{-16}$, $n = 74$), suggesting that marker strength is consistently reported in the data and text.[23] The adipose (Hildreth) dataset illustrates this pattern: reported markers were strong on average, but the best recovery occurred at $N = 61$ for image markers and $N = 41$ for text markers (**Supplementary Figure S1**).[24]

We then asked whether a fixed rank cutoff on the DEG table could substitute for expert marker curation. We treated the top N DEGs as predicted markers, swept N from 1 to 20,000, and compared those predictions

A **LLMarkers benchmark**

Paper	CTs	Genes	Pairs	Tissue
Emont 2022	45	105	346	adipose
Hildreth 2021	31	141	234	adipose
He 2021	27	75	370	bone marrow
Gautam 2021	32	94	251	retina
Adams 2020	20	124	135	lung
Wagner 2020	7	62	58	ovary
Shamis 2020	19	125	171	testis
Total	**168**	**641**	**1,560**	**6 tissues**

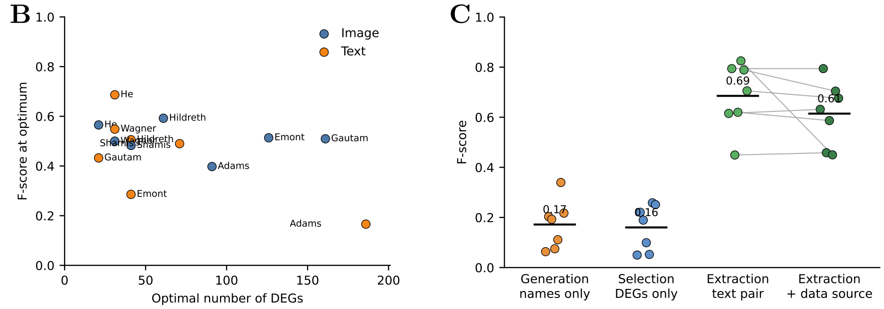

Figure 2: `LLMarkers` **enables marker gene evaluation. A**. Summary of the seven studies in the `LLMarkers` benchmark. **B**. Optimal DEG rank cutoff and F-score for recovering reported markers from text and images. **C**. F-score for LLM-marker generation from cell type names, LLM-marker selection from DEGs, LLM-marker extraction from manuscript text, and the stricter extraction task requiring the correct DEG table. Each point is one benchmark study; black bars show method means. Gray lines connect the same dataset.

to the curated marker set. Across datasets, the optimal N averaged 76 for image markers and 60 for text markers, with peak F-scores of 0.51 and 0.44, respectively (**Figure 2B**).[25] Thus, selecting roughly the top 60–80 DEGs per cell type recovered only about half of the markers authors chose to report. Marker-selection methods commonly rank or optimize genes for discriminating labeled cell groups [Pullin and McCarthy, 2024, Dumitrascu et al., 2021], and DE calls can vary across single-cell methods and filtering choices [Soneson and Robinson, 2018], but DEG rank alone does not replace the biological judgment, prior literature, validation, and reporting choices reflected in the manuscript.

Automatically Curating Marker Genes

Because reported markers could not be recovered by DEG rank alone, we next asked whether LLMs could recover the markers that papers actually report. We evaluated three LLM tasks with increasing access to study-specific context: marker generation from cell type names, marker selection from ranked DEG tables, and marker extraction from manuscript text (**Figure 2**; **Supplementary Table S2**), building on prior NLP-based marker extraction and language-model approaches to single-cell representation and cell type annotation [Cheng et al., 2024a, Hou and Ji, 2024, Levine et al., 2023, Zhao et al., 2024, Schaefer et al., 2025, Yang et al., 2025]. We also built `mrkr`, a tool that prompts an LLM, structures marker records, and verifies extracted labels and genes against their source text.

Marker generation tested whether LLM knowledge alone could recover reported markers. Given only cell type names, the model generated 1,390 unique cell type–gene pairs, and 845 (61%) matched at least one DEG entry in the corresponding studies.[26] These genes were often plausible markers, but they were usually not the genes authors reported. Generation achieved a mean pair F1 of 0.17 against expert-curated markers. Performance was lowest for study-specific labels such as "ADIPOCYTE 1" through "ADIPOCYTE 7" in `adipose (Emont)` (F1 = 0.06) and "LBM1" through "LBM3" in `bone` (F1 = 0.19).[27] These labels depend on the local experiment. The reverse task gave the same message from the other direction. Given anonymized gene lists, the model predicted the exact reported label for 27 of 181 cell types (15%); in total, 131 of 181 predictions (72%) were exact, substring, or shared-token matches.[28] For example, it predicted "ROUND SPERMATID" for "ROUND SPERMATID CELL." Gene lists often carried broad cell type signal, but the reported label did not specify the exact marker program chosen in a paper.

Marker selection tested whether LLMs could use study-specific DEG tables to choose reported markers. We gave the model the top N DEGs per cell type and asked it to choose markers from the ranked list. The model mostly selected genes from the supplied tables, but mean pair F1 was 0.16 at each dataset's optimal N, no better than generation.[29] Even under the DEG-constrained maximum, selection reached only 20.2% of the attainable F1 on average.[30] This result is consistent with our prior analysis that DEG tables contain useful evidence, but the reported markers depend on additional information.

Marker extraction performed best because it used the manuscript itself. Using `mrkr`, we extracted 392 total records, representing 357 within-dataset unique mapped cell type–gene pairs across the seven benchmark studies. Every extracted source sentence matched the manuscript verbatim, and 366 of 392 records (93%) had both the cell type and gene label confirmed within the sentence.[31] When evaluated against text-sourced human annotations, extraction achieved mean pair F1 of 0.69, with precision 0.71 and recall 0.72.[32] Requiring that the model select the correct DEG table for each marker as well as the cell type and gene slightly lowered mean data-level F1 to 0.61.[33]

The main limitations with LLM usage were nomenclature and reporting format. Extraction was evaluated on manuscript text, so it could not recover markers reported only in figures. When evaluated against all human annotations, including figure-only markers, extraction pair F1 fell to 0.32.[34] Errors also reflected common reporting problems, including abbreviations such as "SMCs," gene-range shorthand such as "HOX9–11," and labels that were semantically correct but not verbatim matches to the source sentence. Still, among the three tasks, extraction was the only one that recovered paper-specific marker claims while preserving source evidence. It cost \$3.77 total, or \$0.54 per paper, compared with \$2.30 for generation and \$5.94 for selection.[35]

Comparing Labels and Markers

Having shown that extraction can recover paper-specific marker claims, we used it to ask whether locally reported markers transfer across studies, a problem closely related to prior work on cross-dataset cell-type replicability [Crow et al., 2018]. We ran `mrkr` on 504 bioRxiv preprints and 457 Human Cell Atlas-linked publications. Across these corpora, `mrkr` extracted 32,621 marker associations from 855 papers that yielded marker records. The supporting text matched the manuscript exactly, and both the cell type label and gene label were confirmed within the extracted sentence for 89% of bioRxiv records and 93% of HCA-linked records.[37] For cross-study analysis, we kept verified human markers that mapped to Ensembl gene IDs and deduplicated records within each paper. This produced 24,141 analysis-ready marker records and 3,351 paper-level marker profiles with at least three mapped marker genes, where each profile is a reported cell type label and its mapped marker-gene set in one paper.[38]

We then compared every cross-paper cell-type profile by label similarity and marker gene similarity (**Figure 3A**). If labels were self-contained definitions of cell types, then profiles with the same label should usually report the same marker genes. They did not. Among profile pairs with the exact same normalized label, only 45 had identical marker sets, while 2,408 had partial overlap and 3,167 had no shared marker gene. Thus, 99.2% of same-label pairs did not report the same marker set.[39] The reverse also occurred. Across profiles with different labels, 102,513 pairs shared at least one marker gene.[40] Reported labels therefore carry useful information, but they do not specify a marker-gene set on their own.

This disagreement was clearest for recurrent immune cell types (**Figure 3B**). Among cross-paper profile pairs with the same label, 54% of "T cell" pairs, 64% of "CD8 T cell" pairs, 65% of "CD4 T cell" pairs, 70% of "macrophage" pairs, and 74% of "monocyte" pairs shared no marker gene.[41] The monocyte examples in **Supplementary Table S3** show the same-name case in more detail. Different labels also pointed to the same marker program. "B cell" and "B cells" shared *CD79A*, *MS4A1*, and *CD19*; "monocyte" and "monocytes" shared *S100A8*, *S100A9*, and *CD14*; and "naive T cell" and "TCM" shared *TCF7*, *CCR7*, *LEF1*, and *SELL*.[42] Reviewed exact marker-profile neighborhoods are summarized in **Supplementary Table S4**. These cases show that a reported label does not state which marker program, cell state, subtype, or local comparison it denotes.

We observed the same pattern in an independently curated resource (**Figure 3D,E**). Among the HCA's Cell Annotation Platform (CAP) human profile pairs from different projects with the exact same reported label, only 2 of 307 pairs (0.7%) had identical marker sets, while 97 pairs (31.6%) shared no marker gene.[43] Same-label CAP profiles shared markers more often than random different-label pairs, but the overlap was still incomplete. Across 307 cross-project same-label CAP profile pairs, 68% shared at least one marker gene and the mean marker-set Jaccard was 0.161, compared with 2% and 0.003 for random different-label pairs.[44] Thus, the label-marker mismatch was not specific to our extraction pipeline.

Identifying Global Markers

The pairwise comparison creates an apparent tension. Same-label profiles often did not share marker genes, yet cell type labels are not arbitrary. If authors use a shared marker vocabulary, then local marker sets may disagree pairwise while still recovering common markers in aggregate.

We tested this by measuring local specificity and global recovery for recurrent labels (**Figure 3C**). For each paper-level profile, we first identified markers that were specific within the source paper, meaning genes not reported as markers for other labels in that paper. We then asked what fraction of those locally specific markers appeared in any same-label profile from another paper. This measure does not require two papers to report identical marker sets. It asks whether the local marker vocabulary for a label is recoverable across other studies in the corpus.

Many immune labels showed both high local specificity and substantial global recovery. "T cell" had mean local specificity 0.90 and mean global recovery 0.80, "B cell" had 0.91 and 0.70, "TREG" had 0.73 and 0.69, "macrophage" had 0.93 and 0.60, and "monocyte" had 0.88 and 0.46.[45] Same-label profiles also shared marker genes more often than random profile pairs, but the distribution remained broad (**Figure 3B**). Across 5,620 cross-paper same-label LLMarkers profile pairs, 44% shared at least one marker gene and the mean marker-set Jaccard was 0.098, compared with 2% and 0.003 for random different-label pairs. The median same-label Jaccard was still 0.[46] These values explain why the pairwise results looked discordant. Individual immune profiles often report different marker subsets, but same-label profiles still recover shared marker vocabulary when aggregated across studies.

Because CAP records Cell Ontology terms for cell annotations [Osumi-Sutherland et al., 2021], we also

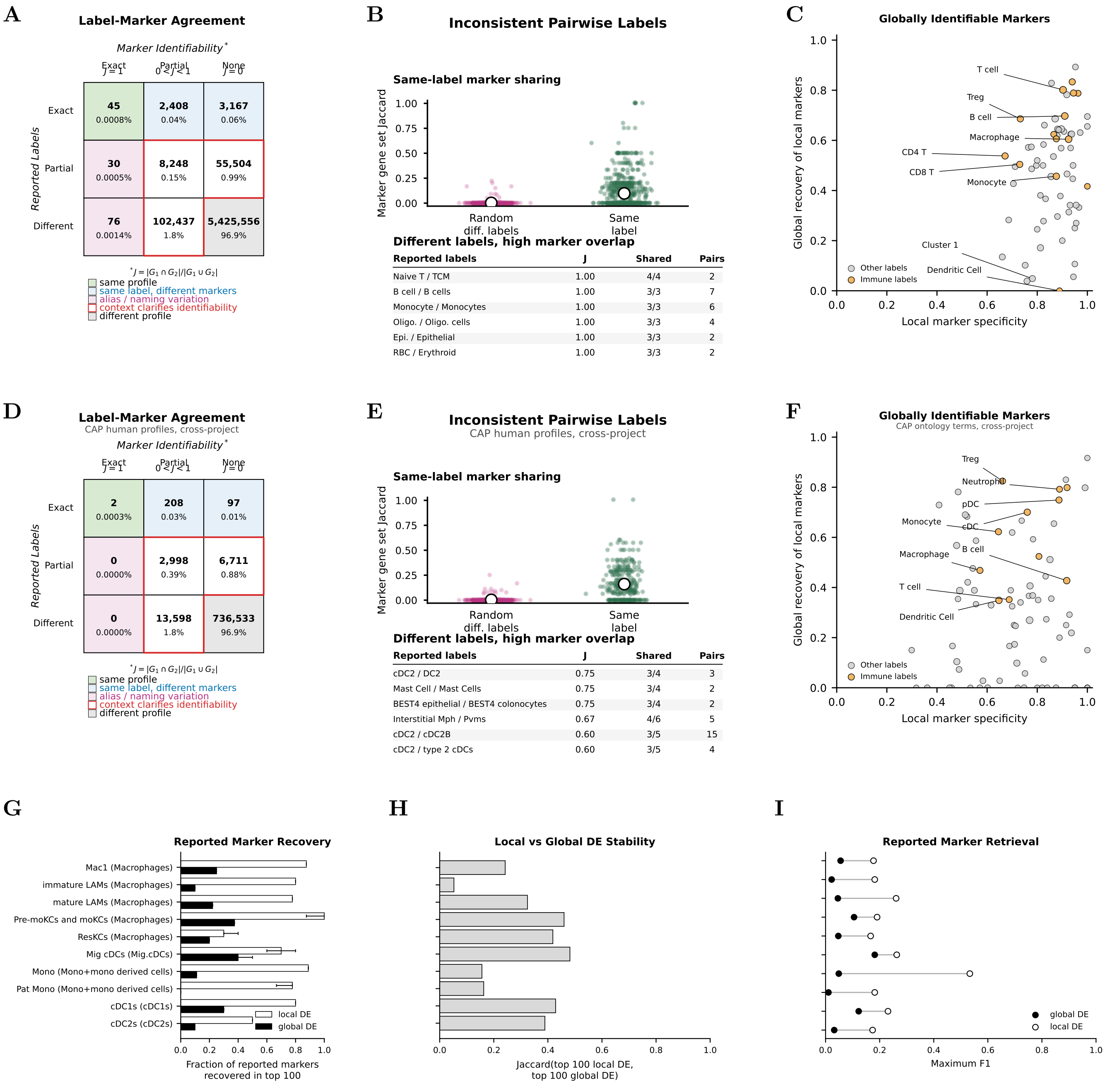

Figure 3: Corpus-scale evaluation of marker genes and cell-type labels. A–C. LLMarkers extracted corpus. **D–F**. CAP human marker profiles, restricted to cross-project comparisons. **G–I**. CAP liver expression-data example comparing a local myeloid dataset to the broader all-cells liver atlas. **A,D**. Label-marker agreement. Counts are cell-type pairs, grouped by reported label similarity and marker-gene set similarity. **B,E**. Inconsistent pairwise cell type labels. Swarm plots compare marker-gene set Jaccard for random different-label profile pairs and same-label profile pairs. Tables list different reported labels with high marker overlap. **C,F**. Globally identifiable markers. Each point is a recurrent label or ontology term. The x-axis is the mean fraction of markers that are unique to a profile within its source paper or CAP project. The y-axis is the mean fraction of those locally specific markers recovered anywhere in same-label or same-ontology-term profiles from other papers or projects. Immune labels are highlighted in orange. **G–I**. Reported markers were compared with one-vs-rest DE rankings in the local myeloid dataset and after mapping those labels to the broader all-cells liver context. Whiskers in **G** show 95% intervals from 50 stratified cell-subsampling replicates.

grouped CAP profiles by ontology term rather than reported label (**Figure 3F**). Across 78 recurrent CAP ontology terms, median local specificity was 0.72 and median global recovery was 0.33; "B cell" had mean local specificity 0.92 and mean global recovery 0.43, while "neutrophil" had 0.89 and 0.79.[47] Ontology labels reduce nomenclature variation and recover marker vocabulary above chance, but they do not by themselves make locally reported marker sets globally self-contained.

We then tested the local/global distinction directly in expression data from CAP liver (**Figure 3G–I**). We compared reported markers from a local liver myeloid dataset against one-vs-rest DE rankings in that same dataset and after mapping each local label to its broader label in the all-cells liver atlas. Across 10 local myeloid labels, local DE recovered 74% of reported markers in the top 100 on average, while global DE recovered 21%; every local label had higher local than global recovery. The mean maximum F1 for reported marker retrieval was 0.24 locally and 0.07 globally.[48] Reported markers therefore behaved as local features of the partition that produced them. Some lifted into the broader atlas context, but many did not.

Together with the DEG-rank analysis, this suggests that marker reporting is not a simple readout of the strongest genes in each dataset. Authors draw from shared field knowledge, but instantiate different subsets of that knowledge depending on tissue, disease, assay, and the local partition being annotated. The label is useful because it anchors aggregation. It is insufficient because it does not specify which local evidence produced the marker set.

Defining a Marker Gene and Cell Type

The `LLMarkers` and CAP analyses show why a marker gene needs context. A gene can be useful for naming a cluster in one study, yet fail when the same label is compared across studies or placed into a broader atlas. This motivates a definition for a marker gene and cell type that takes into account the study s, partition p, target group t, and comparison family C (Supplementary Note). This requirement is consistent with the view that single-cell data can define cell types and states only relative to explicit criteria [Trapnell, 2015, Zeng, 2022]. We encode marker-gene evidence as binary set membership, matching marker-matrix or marker-set annotation methods [Pliner et al., 2019, Zhang et al., 2019, Ianevski et al., 2022, Bi et al., 2026] and transcript-presence enrichment assays [Abay et al., 2024], and assume that marker discovery is implemented through between-cluster comparisons using the same differential-expression machinery that underlies standard marker selection practice [Robinson et al., 2009, Pullin and McCarthy, 2024].

Let $M_{s,p}(t, h)$ be the genes called as markers for cell-type t against comparison cells h. Then the marker set for t relative to C (the set of all comparison cell groups) is

$$M_{s,p}(t; C) = \bigcap_{h \in C} M_{s,p}(t, h). \tag{1}$$

In words, a gene marks a cell type only if it distinguishes that cell type from all sets of cells in the chosen comparison family.

The formalization shows that cell-type marker sets are intersections of pairwise marker sets, and that broadening the comparison family can only remove markers.[49] Thus, in single-cell RNA-seq, a cell type label is not a self-contained definition of marker identity. It is a measurement statement tied to the cells assayed, the partition analyzed, and the alternatives from which the target group was distinguished. This turns the empirical pattern in `LLMarkers` and CAP into a reporting requirement: reusable cell-type definitions need machine-readable marker records that retain study, partition, target group, comparison family, and supporting evidence. One-versus-all marker lists and flat cell type–gene entries compress away that context, whereas pairwise marker calls preserve the information needed to rebuild local, hierarchical, or atlas-level marker sets. Recomputing pooled one-versus-all statistical tests, such as t-tests, additionally requires sufficient statistics such as group means, variances, and sample sizes (Supplementary Note). The same inclusion result is consistent with tree-based views of hierarchical cell type relationships and hierarchical immune-cell annotation [Domcke and Shendure, 2023, Beltz et al., 2026]: a parent cell type marker remains valid in a child cell type, but a child cell type marker need not remain valid in the parent or global comparison family.

The `LLMarkersDB` Enables Marker Search

Given the importance of context to define a cell type marker gene, we compiled the provenance-linked marker records into `LLMarkersDB`. The database contains 10,132 paper-level marker profiles, including 183 profiles from the human-curated benchmark, 3,805 profiles from the bioRxiv corpus, and 6,144 profiles from Human Cell Atlas-linked publications.[50] Each profile stores a reported cell type label, mapped marker genes, and the quoted manuscript text supporting the claim. `LLMarkersDB` supports free-text queries, such as "cytotoxic lymphocytes" or "stromal fibroblasts," using text embeddings, and gene-set queries using Jaccard similarity over marker genes.[51] This lets a user query a label, context, or gene set and recover the source papers and supporting sentences behind matched marker claims. The database is available at https://sina.bio/llmarkers.

3 Discussion

Marker genes are often treated as portable definitions of cell types. Our results argue for a narrower interpretation. Reported marker genes are local evidence: they record that a gene helped identify a population in a particular experiment, under a particular comparison, interpreted through prior knowledge. That distinction matters for atlas building, because atlas-scale organization requires trackable labels, taxonomies, and cell-type relationships across studies [Miller et al., 2020, Domcke and Shendure, 2023]. Our results show that most papers report the compressed marker claim without the comparison that made it meaningful. This reframes the role of cell type labels. Labels are not arbitrary, and recurrent immune labels in our corpus recovered shared marker vocabularies in aggregate. But the same label often pointed to different reported marker sets, and different labels sometimes pointed to the same marker program. This is not simply the nomenclature problem described by common naming systems [Miller et al., 2020] or the broader cell-type definition and terminology problem [Zeng, 2022, Zhu et al., 2026]. Authors assign labels by comparing local gene programs to prior label–gene associations, so in our data the local marker problem and the naming problem are coupled in practice.

Improving global marker identifiability will require preserving more of the local comparison (Supplementary Note). One-versus-all differential expression is useful within a study, but binary DEG calls can be sensitive to method choice, filtering, thresholds, replicate structure, and biological or technical variation [Soneson and Robinson, 2018, Squair et al., 2021, Hoerbst et al., 2025], and even large single-cell cohorts can show incomplete DEG replication within the same cell type [Krzak et al., 2026]. Pairwise comparisons, atlas-level reanalysis, and better machine-readable reporting of cell-type claims [Lubiana et al., 2022] could expose distinctions that current marker lists hide. Cell ontologies remain essential, but ontology terms are most useful when they remain linked to source annotations, measurements, and reference evidence [Osumi-Sutherland et al., 2021, Tan et al., 2026].

LLMs were useful here because they helped recover that connection to the underlying evidence. Their value was not in replacing expert knowledge with generated marker lists. LLM marker generation produced plausible markers, but extraction from manuscripts recovered paper-specific claims more faithfully and kept those claims linked to source text. Related LLM approaches use expression-derived inputs or transcriptome-text pretraining for cell-type annotation and data exploration [Hou and Ji, 2024, Levine et al., 2023, Zhao et al., 2024, Schaefer et al., 2025, Yang et al., 2025]; here, the practical role for LLMs was marker curation, where they organized human-reported evidence so that it could be queried, compared, and checked across studies.

The same issue likely extends to single-cell foundation models evaluated on cell-type separation or classification [Kedzierska et al., 2025]. While we did not evaluate those models here, label-based evaluations likely inherit the same local-to-global and nomenclature problems. A benchmark that stores only cell type names can hide whether a model recovered a local marker program, a broad ontology term, or a context-specific state.

A Supplementary Material

Table S1: LLMarkers benchmark. The number of unique cell types (CTs), genes (Ensembl IDs), and (cell type, gene) pairs Overall, and across Text, Image, and DEG tables. The %DEG reports the percentage of pairs found in the study's DEG lists.

	Overall			Text				Image				DEG
Dataset	CTs	Genes	Pairs	CTs	Genes	Pairs	DEG%	CTs	Genes	Pairs	DEG%	Pairs
`Adipose (Emont)`	45	105	346	20	23	30	80	44	104	344	39	43,471
`Adipose (Hildreth)`	31	141	234	21	53	66	92	31	124	207	96	46,357
`Bone (He)`	27	75	370	26	69	98	53	24	29	309	13	5,382
`Eye (Gautam)`	32	94	251	12	22	24	88	28	82	235	49	24,998
`Lung (Adams)`	20	124	135	6	33	33	97	20	99	109	97	34,760
`Ovary (Wagner)`	7	62	58	7	46	37	89	6	54	41	80	14,357
`Testis (Shamis)`	19	125	171	15	91	123	78	11	59	64	94	9,663
Total	168	641	1,560	100	305	409	78	151	495	1,304	53	169,687

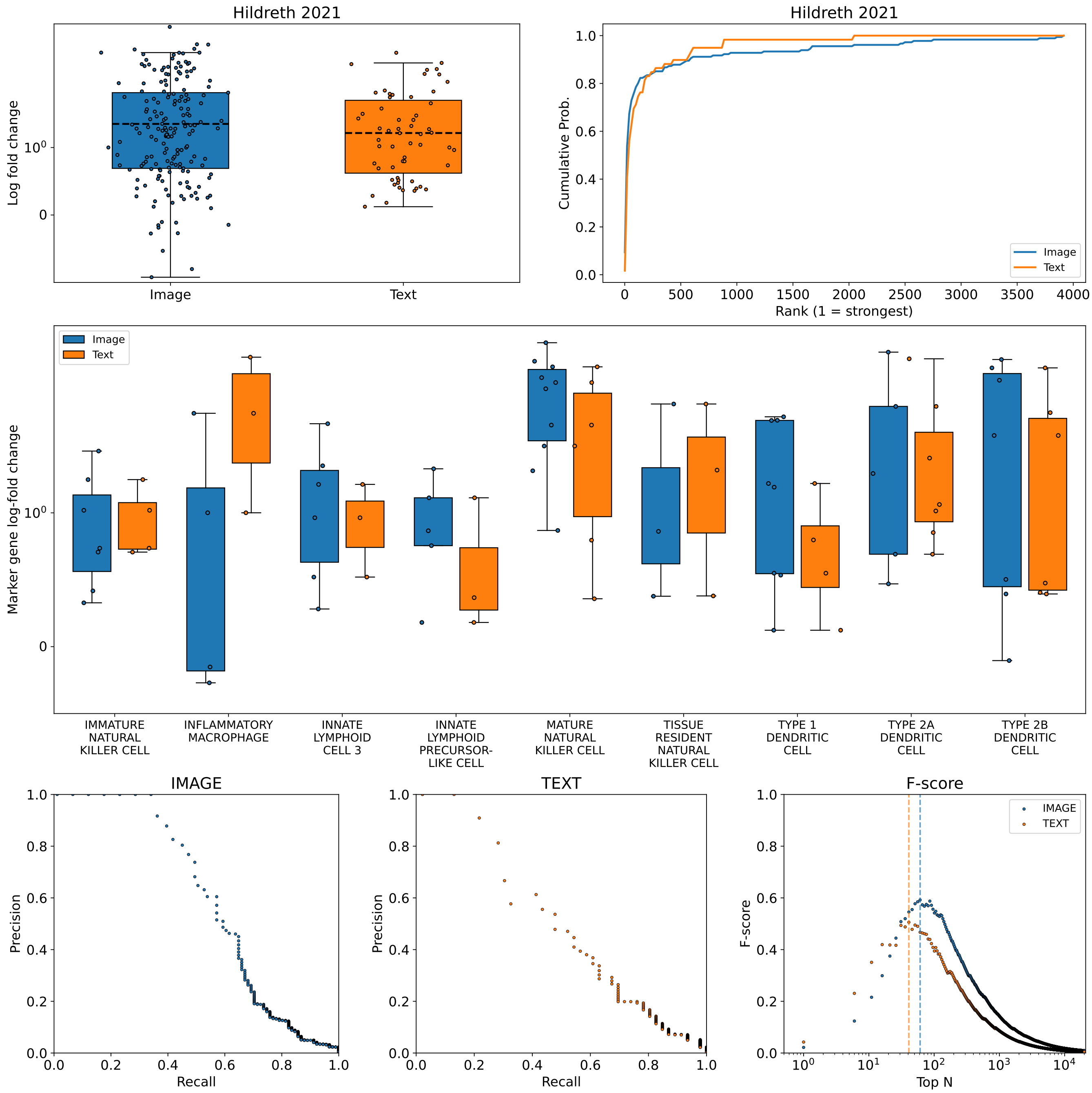

Figure S1: Marker gene selection strength for [Hildreth et al., 2021]. Top row shows log fold change distributions (left) and cumulative rank distributions (right) for discussed markers from images (blue) and text (orange). Middle row shows per-cell-type log-fold change distributions for image (blue) and text (orange) discussed markers. Bottom row shows precision–recall curves for image and text markers, and F-score as a function of the top-N DEG threshold.

Table S2: LLM marker gene curation. All runs used Claude (claude-opus-4-6). P, R, and F1 are precision, recall, and F1 on (group_name, feature_id) pairs evaluated against text-sourced human annotations. N: number of unique mapped pairs produced. %DEG: percentage of produced triples found in the matching DEG table (by data_id). Rank: median DEG rank of matched triples. Opt N: DEG rank threshold maximizing F-score for recovering the produced pair set. DEG columns require a data_id link and are not available for generation (—). Cost: total API cost in USD. Time: wall-clock minutes. Bottom row shows means across datasets. **(a)** Generation. The model was given cell type names only. Pred. Acc.: fraction of cell types correctly identified from anonymized gene lists. **(b)** Selection. The model was given the top DEGs per cell type. Best N: number of top DEGs maximizing pair F1 per dataset. **(c)** Extraction. The model was given the manuscript text. Verified: fraction of extractions where source rationale, group label, and feature label were all confirmed.

(a) Generation

Dataset	Pred. %	N	%DEG	Rank	Opt N	P	R	F1	Cost ($)	Time (min)
`Adipose (Emont)`	13	317	–	–	–	0.03	0.37	0.06	0.48	3.2
`Adipose (Hildreth)`	10	252	–	–	–	0.13	0.48	0.20	0.41	2.8
`Bone (He)`	4	203	–	–	–	0.14	0.30	0.19	0.34	2.3
`Eye (Gautam)`	13	244	–	–	–	0.04	0.42	0.07	0.39	2.6
`Lung (Adams)`	40	165	–	–	–	0.07	0.33	0.11	0.29	2.0
`Ovary (Wagner)`	29	69	–	–	–	0.26	0.49	0.34	0.12	0.8
`Testis (Shamis)`	16	144	–	–	–	0.20	0.24	0.22	0.25	1.7
Mean	18	199	–	–	–	0.13	0.37	0.17	0.33	2.2

(b) Selection

Dataset	Best N	N	%DEG	Rank	Opt N	P	R	F1	Cost ($)	Time (min)
`Adipose (Emont)`	200	371	97	17	106	0.03	0.33	0.05	1.53	4.4
`Adipose (Hildreth)`	400	333	98	20	106	0.13	0.67	0.22	0.50	4.2
`Bone (He)`	300	293	97	26	116	0.13	0.38	0.19	0.95	3.4
`Eye (Gautam)`	20	218	95	8	21	0.06	0.50	0.10	0.43	2.7
`Lung (Adams)`	200	349	97	53	196	0.03	0.30	0.05	1.24	3.9
`Ovary (Wagner)`	200	110	100	28	76	0.17	0.51	0.26	0.35	1.3
`Testis (Shamis)`	300	227	99	32	181	0.19	0.36	0.25	0.93	2.8
Mean		272	98	26	115	0.11	0.44	0.16	0.85	3.2

(c) Extraction

Dataset	Verified	N	%DEG	Rank	Opt N	P	R	F1	Cost ($)	Time (min)
`Adipose (Emont)`	100%	36	75	216	51	0.57	0.67	0.62	0.29	1.3
`Adipose (Hildreth)`	97%	71	87	112	56	0.77	0.82	0.79	0.57	2.6
`Bone (He)`	98%	40	57	28	36	0.78	0.32	0.45	0.81	3.7
`Eye (Gautam)`	90%	48	69	269	21	0.47	0.92	0.62	0.38	1.7
`Lung (Adams)`	85%	50	72	666	186	0.70	1.00	0.82	0.55	1.9
`Ovary (Wagner)`	97%	35	100	72	31	0.82	0.76	0.79	0.28	0.9
`Testis (Shamis)`	91%	85	77	55	71	0.87	0.59	0.71	0.88	3.2
Mean	94%	52	77	203	65	0.71	0.72	0.69	0.54	2.2

Table S3: Different marker genes. Monocyte marker profiles across 29 bioRxiv preprints, restricted to mapped human markers with at least three genes per profile. For papers with multiple monocyte-labeled profiles, we retained the cell type with the most extracted markers (one row per paper). The context column gives a brief study setting (for example tissue, disease, or analysis type). Across these 29 profiles, mean pairwise Jaccard similarity is 0.063, showing strong cross-study heterogeneity.

Paper	Cell type label	Context	Extracted monocyte markers
[Preglej et al., 2025]	MONOCYTE	Rheumatoid arthritis	CXCL8, EGR1, FCN1, JAK2, NFKBIZ, TNFAIP3
[Urb, 2025]	ISG+ AP1LOW CLASSICAL MONOCYTES	Sjogren's disease PBMC	IFI44L, IFI6, ISG15, MX1, MX2
[Bachurski et al., 2025]	MONOCYTE	EV immunomodulation assay	BCL2, CD14, CD25, CD274, IL3RA, IL7R
[Imoto, 2025]	CD14+ MONOCYTE	Method benchmark dataset	CD14, LGALS3, LYZ, S100A8
[Wang et al., 2020]	CD14+ MONOCYTE	Method paper (scRNA)	CD14, S100A12, S100A8, S100A9, VCAN
[Wu2, 2022]	MONOCYTES/ MACROPHAGES	Azoospermia testis	APOE, DCN, TIMP3
[Shangguan et al., 2021]	MONOCYTE	HIV vaccine response	CD68, CTSD, GAA, IL17RA, SEMA4A
[Zhao and Fang, 2024]	CD14+ MONOCYTE	Type 2 diabetes PBMC	CD14, CD163, FCN1, LGALS3, MS4A6A, S100A8, S100A9, VCAN
[Obolenska et al., 2025]	MONOCYTE	Placenta pregnancy	LAIR1, S100A8, S100A9
[Rigby et al., 2023]	CLASSICAL MONOCYTE	Type I interferon response	CCL2, CCL8, CD169, CXCL10, IFI27, IFI44, IFITM3, OAS2, OAS3, OASL
[Mar, 2020]	MONOCYTE PROGENITORS AND MONOCYTES	Developmental haematopoiesis	CD14, CD33, MPEG1
[Asaba et al., 2024]	MONOCYTE-MACROPHAGES	COVID-19 BAL fluid	AIM2, APOC1, AVP, BAX, CCL2, CCL3, CD274, CXCL8, FABP4, FOS, GSDMD, IL-6, JUN, MAFB, MARCO, MCL1, MRC1, MSR1, NFE2L2, NRP1, SERPINA1, TIMP1, TLR2, TLR4, TREM2, TREX1, VEGFA
[Gavriilidis et al., 2024]	MONOCYTE	Severe COVID-19	CKAP4, NCF2, PTEN
[Lawlor et al., 2020]	MONOCYTE	Stimulated PBMCs	CCL3, CCL4, CXCL8, IL-6, IL1B, MT1F, MT2A, S100A8, S100A9
[Satpathy et al., 2019]	MONOCYTE	Immune chromatin atlas	CD14, CSF1R, TREML4
[Sherwood et al., 2024]	MONOCYTES/ MACROPHAGES	Zika brain-tumor model	CCR1, CSF2RA, IFNAR1, IFNAR2, TNFRSF1A, TNFRSF1B
[Shemonti et al., 2025]	MONOCYTE	Flow cytometry method	CD34, CD38, CD45, KIT
[Xu, 2025]	MONOCYTE	Pancreatic cancer (PDAC)	FLNB, HABP4, KCNN3
[Bal, 2025]	CD14+ MONOCYTE	Atopic dermatitis	ARHGEF10, ARL17B, BMP6, ERICH1-AS1, GALNT13, HBB, IFI27, LINC00176, NEBL, PDE4C, PDK4, SLC35F1, ZNF727
[Alkhadrawi et al., 2025]	MONOCYTE	Feature selection benchmark	CD14, LYZ, S100A9
[Yu et al., 2024]	MONOCYTE	Disease genetics atlas	CD14, S100A8, S100A9
[Wang et al., 2025]	MONOCYTE	Intrahepatic cholangiocarcinoma	FCN1, S100A8, S100A9
[Zha, 2022]	CLASSICAL MONOCYTES	Checkpoint inhibitor toxicity	B2M, CD38, IL15, STAT1
[Li et al., 2021]	MONOCYTE	Acute myeloid leukemia	CD52, CLCN5, HAL, ICAM3
[Luo, 2025]	CD4+ MONOCYTE	CAR-T solid tumors	CD14, CD16, IL1B, PFKFB4, TNF, TNFSF13
[Yang et al., 2019]	MONOCYTE	Ambient RNA correction	CD14, LYZ, S100A8, S100A9
[Jeo, 2025]	DYSFUNCTIONAL CD14 MONOCYTE	ML method benchmark	S100A12, S100A6, S100A8, S100A9, TSPO
[Cheng et al., 2024b]	MONOCYTE-DERIVED CELL	Glioblastoma microenvironment	S100A8, S100A9, VCAN
[Lea, 2020]	LN MONOCYTES	Lung and lymph node	CD14, F13A1, FOLR2, LYVE1, MAF, MAFB, MMP9, S100A8, S100A9, STAB1, TIMD4

Table S4: Similar marker genes. Exact marker-profile neighborhoods from the cross-study marker graph. Nodes are paper-level marker profiles, and profiles were connected when they came from different papers and shared the same mapped human marker genes ($J = 1$). Each block shows one shared marker profile, the reported cell type labels attached to that profile, and the source context. The table reports label–marker separation patterns rather than biological classifications.

Paper	Reported cell type	Context	Label–marker separation	Marker profile
Shared marker profile 1: TCF7, CCR7, LEF1, SELL				
[Ma et al., 2020]	NAÏVE T CELL	Liver cancer	Different labels; identical markers	TCF7, CCR7, LEF1, SELL
[Kim et al., 2024]	TN	Lung cancer immune-checkpoint study	Different labels; identical markers	TCF7, CCR7, LEF1, SELL
[Kim et al., 2024]	TCM	Lung cancer immune-checkpoint study	Different labels; identical markers	TCF7, CCR7, LEF1, SELL
[Yoffe et al., 2025]	TREG.1	Preinvasive lung cancer	Different labels; identical markers	TCF7, CCR7, LEF1, SELL
[Boland et al., 2020]	NAÏVE/MEMORY	Ulcerative colitis	Different labels; identical markers	TCF7, CCR7, LEF1, SELL
Shared marker profile 2: S100A8, S100A9, S100A12				
[Conde et al., 2022]	CLASSICAL	Cross-tissue immune atlas	Different labels; identical markers	S100A8, S100A9, S100A12
[Werba et al., 2023]	MDSC	Pancreatic cancer after chemotherapy	Different labels; identical markers	S100A8, S100A9, S100A12
[Pelka et al., 2021]	MACROPHAGE	Colorectal cancer immune hubs	Different labels; identical markers	S100A8, S100A9, S100A12
Shared marker profile 3: C1QB, C1QC, CD163				
[Liu et al., 2022]	MAC3	Chronic inflammatory skin disease	Different labels; identical markers	C1QB, C1QC, CD163
[Kong et al., 2023]	MACROPHAGE	Crohn disease immune atlas	Different labels; identical markers	C1QB, C1QC, CD163

B Methods

B.1 Benchmark construction and verification

The LLMarkers benchmark was assembled by manual curation of seven single-cell RNA-seq studies spanning six tissues: `adipose` [Emont et al., 2022, Hildreth et al., 2021], `bone` [Zhong et al., 2020], `eye` [Gautam et al., 2021], `lung` [Adams et al., 2020], `ovary` [Wagner et al., 2020], and `testis` [Shami et al., 2020]. The benchmark was designed as a carefully linked evaluation set rather than a comprehensive survey of single-cell reporting styles. For each study, a curator read the manuscript and recorded every marker gene–cell type association, the verbatim sentence or figure panel from which it was drawn (source_rationale), whether it came from text or a figure (source_type), the supplementary data source containing the corresponding differential expression results (data_id), and the gene's Ensembl identifier (feature_id). Gene symbols were mapped to Ensembl IDs using `mrkr map`, which queries the Ensembl REST API.[57]

Each study's supplementary tables were parsed to extract differential expression gene lists. Header detection was performed automatically by scanning the first 10 rows for a row containing expected column names (e.g., gene, p_value, avg_log2FC). The resulting DEG lists were stored in the same evidence schema as human annotations, enabling direct comparison.[58]

To verify the fidelity of the human annotations, we checked each source_rationale against the normalized manuscript text using exact substring matching after whitespace and case normalization. Across all seven datasets, 96.2% of source rationales matched the manuscript exactly. We additionally verified that the group label (cell type name) and feature label (gene symbol) could be localized within each source rationale using the `taln` text alignment library, which performs subword-level alignment. Group labels were confirmed in 97.5% of records and feature labels in 99.5%.[59]

B.2 Marker strength analysis

To quantify where human-curated marker genes fall in the differential expression ranking, we matched each marker to its corresponding entry in the dataset's DEG list. For human-curated markers, matching was performed on the four-part key (dataset, data_id, group_name, feature_id), requiring exact agreement on the cell type, gene, and supplementary data source. For LLM-generated markers, which have no data_id, we matched on (group_name, feature_id) against all DEG data sources within each dataset. Markers that could not be matched—because the gene was absent from the DEG list or lacked an Ensembl ID—were excluded from this analysis.[60]

For each matched marker, we recorded its rank within the cell type's DEG list (rank 1 = highest log-fold change) and its log-fold change. We computed the fraction of markers falling within the top N DEGs for thresholds $N \in \{10, 25, 50, 100\}$ and separately tabulated these fractions for text-sourced markers, image-sourced markers, and generated markers.[61]

To test whether discussed markers are drawn from stronger DEGs than expected by chance, we performed permutation tests at the level of individual cell types within each dataset and data source. For each cell type with at least two discussed markers, we sampled a size-matched set of genes from the top 100 significant ($p < 0.05$) DEGs that were not in the human-curated marker set. We then compared the mean rank of the discussed markers against the null distribution obtained by randomly partitioning the combined set 10,000 times. Tests were two-sided at $\alpha = 0.05$.[62]

To estimate the optimal DEG list depth for recovering each marker set, we computed precision–recall curves by varying a rank threshold N from 1 to 20,000. At each threshold, we defined the top N DEGs as predicted markers and computed precision, recall, and the F-score. The optimal N was defined as the threshold that maximized the F-score. This analysis was performed separately for each dataset and marker source (text, image, or generated).[63]

Differences in rank and log-fold change between marker sources were assessed using two-sided Mann–Whitney U tests.[64]

B.3 LLM generation and extraction

We evaluated three approaches to LLM-based marker gene curation across the seven benchmark datasets. All experiments used Claude (claude-opus-4-6) with temperature 0 and a maximum output length of 32,000 tokens; we did not benchmark model-to-model variation. Costs were computed from the model's API pricing ($5 per million input tokens, $25 per million output tokens for claude-opus-4-6).[65]

In the generation experiment (celltypes-to-genes), we gave the model the list of cell type names from each dataset's human-curated marker set and asked it to generate 5–15 well-established marker genes per cell type. The model received only the species and cell type names—no manuscript text, figures, or supplementary data. In a second generation experiment (genes-to-celltypes), we extracted the gene list for each cell type from the human-curated set, replaced the cell type names with anonymous labels (Group 1, Group 2, . . .), and asked the model to predict the cell type for each group. We measured prediction accuracy as the fraction of exact string matches between predicted and original cell type names. Non-exact predictions were counted as related when they showed substring containment or shared label tokens; remaining predictions were counted as having no lexical relation.[66]

In the selection experiment, we parsed each dataset's DEG tables, ranked genes per cell type by absolute log fold change, and presented the top N genes (with log fold change and adjusted p-value) to the model, asking it to select which are true markers. We swept $N \in \{10, 20, 30, 40, 50, 100, 200, 300, 400, 500\}$ for each dataset and reported the N that maximized pair-level F1. As a control for whether cell type identity aids selection, we repeated the full sweep with cell type names replaced by anonymous cluster labels (Cluster 1, Cluster 2, . . .), preserving the gene lists and statistics but removing biological context from the names.[67]

To quantify the best possible selection performance, we computed a DEG-constrained max F1 score at each N. Let H be the set of human-curated marker pairs and D_N be the set of marker pairs available in the top-N DEG lists shown to the model. The recoverable set is $R_N = H \cap D_N$. Any selector restricted to D_N cannot recover pairs outside R_N. The maximum achievable pair-level F1 is therefore

$$\mathrm{F1}_{\max}(N) = \frac{2|R_N|}{|H| + |R_N|}, \tag{2}$$

which is attained by selecting exactly the recoverable set R_N (precision = 1). In **Figure 2**, the bound markers for selection are plotted at each dataset's selected operating point using that same dataset-specific N.

In the extraction experiment, we ran the `mrkr extract` pipeline on each manuscript, followed by `mrkr verify` and `mrkr map`. We compared the extracted evidence against human curation at three levels of stringency. At the feature level, we computed the Jaccard similarity between unique gene sets (Ensembl gene IDs), ignoring cell type assignment. At the pair level, we defined a match as a (group_name, feature_id) pair present in both sets and computed precision, recall, and F1. At the data level, we required a full (data_id, group_name, feature_id) triple to match. We also computed a mean per-cell-type gene F1 by averaging the gene-level F1 across all cell types shared between the LLM and human sets.[68]

Because the LLM extracts only from manuscript text while human curators also annotated markers from figures and supplementary tables, we evaluated all three methods—generation, selection, and extraction—against the subset of human annotations with source_type = "text" (**Supplementary Table S2**; **Figure 2**). We additionally report extraction metrics against all annotations to quantify the effect of image-only markers. All comparisons used Ensembl gene IDs (feature_id) rather than gene symbols to avoid ambiguity from gene name variants.[69]

Source verification statistics were derived from the `_verification` field attached to each LLM extraction by `mrkr verify`. This field records whether the source rationale was found verbatim in the manuscript text, and whether the group label and feature label could be aligned within the source rationale using the `taln` text alignment library.[70]

B.4 bioRxiv

We selected bioRxiv preprints for marker gene extraction from a local mirror of 428,741 MECA archives downloaded from the bioRxiv source repository (s3://biorxiv-src-monthly/, accessed January 2025). Each MECA archive contains the manuscript's JATS XML, from which we extracted structured metadata including title, abstract, author-provided keywords, DOI, and license.[71]

We filtered manuscripts using three criteria. First, we required a permissive license (CC-BY or CC0) to allow downstream text processing and redistribution. Second, we required evidence of human relevance by matching the abstract or author-provided keywords against terms including "human", "Homo sapiens", "patient", "clinical", and "PBMC". Third, we required relevance to cell-type marker genes by matching against two sources: (i) author-provided JATS keywords, available for 49% of manuscripts, matching terms such as "marker gene", "cell type", "scRNA", "single-cell RNA", "deconvolution", and "cell atlas"; and (ii) abstract text, matching phrases such as "marker gene", "cell-type marker", "cell type annotation", and co-occurrences of "single-cell" or "scRNA-seq" with "marker." Manuscripts were included if they matched on either keywords or abstract text. When multiple versions of the same manuscript existed (identified by shared DOI), we retained only the most recent version. This procedure yielded 504 unique manuscripts.[72]

Each manuscript's JATS XML was converted to markdown using a custom JATS parser that preserves article structure (sections, headings, inline figure captions) while stripping references and HTML markup. We then ran `mrkr extract` on each markdown file using text-only extraction (no figures), followed by `mrkr verify` and `mrkr map`. Papers were processed in parallel (25 concurrent jobs). Papers yielding zero extractions were recorded with empty marker files.[73]

B.5 Human Cell Atlas-linked publications

We compiled a second corpus from publications linked to Human Cell Atlas projects. HCA project metadata were used to collect publication titles, URLs, and DOIs. Records were deduplicated by DOI when available and otherwise by normalized title. This produced 515 HCA-linked publication records, of which 475 had cached markdown manuscript text under `data/hca/manuscripts`. The cached files were normalized to the same plain-text format used for the bioRxiv corpus.[74]

We processed the cached HCA-linked manuscripts with the same extraction workflow used for bioRxiv: `mrkr extract`, followed by `mrkr verify` and `mrkr map`. The run produced marker-output files for 457 publications, of which 434 contained at least one marker record. Papers yielding no marker records were kept in corpus summaries but did not contribute marker profiles. Cross-study analyses used only records that were human, verified against source text, mapped to Ensembl gene IDs, and retained after within-paper deduplication.[75]

B.6 Gene ID mapping

Gene symbols were mapped to Ensembl identifiers using a reference file built from Ensembl GRCh38.114. Canonical gene names were extracted from the Ensembl GTF annotation, and gene synonyms were retrieved from the Ensembl REST API, following the approach used by gget [Luebbert and Pachter, 2023]. The resulting mapping file contains 94,000 unique gene names and synonyms mapping to 40,000 unique Ensembl gene IDs. Lookup is case-insensitive. Only human records (organism = "homo_sapiens") were mapped; non-human records were left without Ensembl identifiers.[76]

B.7 Cell Annotation Platform marker profiles

We constructed an independent set of marker profiles from Cell Annotation Platform marker records in `data/cell_annotation_platform/markers.json`. CAP records were restricted to human records with Ensembl gene IDs, standardized with the same gene-mapping utilities used for `LLMarkers`, and grouped by CAP dataset, project, label set, and group name. A CAP marker profile was retained when the group had at least three mapped marker genes. We stored the author-provided group label, CAP dataset and project identifiers, label-set name, source ID, Cell Ontology term and category annotations, and rationale DOI metadata when present.[77]

For ontology-based CAP analyses, we restricted CAP profiles to records with a non-empty Cell Ontology term and a CL identifier, then used the ontology term rather than the author-reported label for label normalization and cross-project grouping.[78]

B.8 Cross-paper cell type analysis

To compare local and global marker definitions, we constructed one binary marker vector for each (paper, reported cell type label) pair. The vector was the set of mapped Ensembl gene IDs reported for that label in that paper. Profiles with fewer than three mapped marker genes were excluded from cross-paper analyses. All analyses were restricted to records that were human, verified against the source text, mapped to Ensembl gene IDs, and deduplicated within each paper.[79]

Cell type labels were normalized by converting to ASCII, uppercasing, replacing non-alphanumeric characters with spaces, splitting letter-number boundaries, and standardizing spacing. Exact label pairs had identical normalized labels. Partial label pairs had one normalized label contained in the other, or shared at least one non-generic token after removing generic terms such as "cell", "cells", "type", and "cluster". All remaining label pairs were labeled different. This rule is string-based and does not use ontology mapping or semantic similarity.[80]

Marker-gene set similarity was measured with Jaccard similarity, defined as the number of shared Ensembl gene IDs divided by the size of the union of the two marker sets. Marker pairs were assigned to three bins: exact if Jaccard similarity was 1, partial if Jaccard similarity was greater than 0 and less than 1, and none if Jaccard similarity was 0. The joint distributions in **Figure 3A,D** count unordered cross-paper or cross-project profile pairs across the label-relation and marker-relation bins.[81]

For the same-label marker-sharing analysis in **Figure 3B,E**, we computed all cross-paper or cross-project pairwise Jaccard similarities among profiles with the same normalized label. As a background comparison, we sampled 100,000 cross-paper or cross-project profile pairs with different normalized labels and computed the same Jaccard statistic. Plotted points were sampled for display, while means and percentages were computed from the full pair set.[82]

For the local/global recovery analysis in **Figure 3C,F**, we first identified, for each profile, the markers that were specific within the source paper or CAP project. A marker was locally specific if it appeared in that profile and was not reported for any other label in the same paper or project. For each recurrent label or ontology term, we then asked whether those locally specific markers appeared in any same-label or same-term profile from another paper or project. Local specificity is the mean fraction of profile markers that are locally specific. Global recovery is the mean fraction of locally specific markers that are recovered in at least one same-label or same-term profile from another paper or project. LLMarkers labels were included when they appeared in at least five profiles. CAP ontology terms were included when they appeared in at least three profiles.[83]

For the different-label examples in **Figure 3B,E**, we searched cross-paper or cross-project pairs with different normalized reported labels and at least three shared marker genes. We grouped candidate pairs by label pair and retained examples with repeated evidence across at least two profile pairs.[84]

For the CAP liver expression-data analysis in **Figure 3G–I**, we used CAP dataset 1440 as the local liver myeloid context and CAP dataset 1437 as the broader all-cells liver context. We loaded each AnnData file with `anndata.read_h5ad`, used the `user_provided` expression layer when present, and used `author_cell_type` as the cell type label. For each local myeloid label, we computed one-vs-rest DE scores for all genes as the difference in mean expression divided by its standard error. We then mapped local labels to their broader all-cells labels using the CAP author labels and compared the local one-vs-rest ranking to the corresponding global one-vs-rest ranking. Reported markers were taken from the CAP marker JSON, restricted to human Ensembl-mapped genes. Top-100 recovery is the fraction of reported markers appearing in the top 100 genes by DE score. For **Figure 3G**, we recomputed the DE rankings across 50 cell-type-stratified subsampling replicates and plotted 2.5th–97.5th percentile intervals. Each replicate sampled 80% of cells within each `author_cell_type`, capped at 10,000 cells per cell type. DE stability is the Jaccard similarity between the top 100 local DE genes and the top 100 global DE genes. Marker retrieval F1 was computed over the ranked DE list by treating reported markers as positives and reporting the maximum F1 over rank cutoffs.[85]

The examples in **Figure 1** were selected from the extracted corpus to illustrate the label and marker relations used in the cross-paper analysis. The toy matrices are schematic. The displayed myeloid marker profiles are real extracted profiles, and their joint distribution was computed only over the displayed profiles using the same label-relation and marker-relation rules described above.[86]

For **Supplementary Table S3**, we selected bioRxiv profiles whose normalized reported label contained "monocyte", retained profiles with at least three mapped marker genes, and kept one profile per paper by choosing the monocyte-labeled profile with the most extracted markers. For **Supplementary Table S4**, we searched cross-paper profile pairs with different reported labels and identical mapped marker sets. We then selected representative exact marker-profile neighborhoods and report the source paper, reported label, source context, and shared marker set. The table reports observed label–marker separation patterns rather than biological interpretations or formal ontology mappings.[87]

B.9 Lean formalization

The formal results were encoded in Lean 4 under `analysis/formal`. The basic model treats reported marker evidence as a binary marker matrix over finite cell-type, profile, or context-conditioned groups and marker genes. A marker panel separates a comparison family when every distinct pair of groups has different binary signatures on at least one selected gene. The contrast-level model additionally records study, partition, target group, comparison family, gene, threshold, pairwise contrasts, background contrasts, and real-valued marker scores. We used `span` to align supplementary-note statements with Lean declarations and inspect the paper-facing dependency graph [Pachter, 2026].[88]

The Lean development proves the marker-family intersection theorem, monotonicity of marker sets under comparison-family restriction, local/global and hierarchy inclusion results, one-vs-all equivalence at the binary pairwise level, weighted-background contrast identities, score-thresholded marker-family results, and examples showing that flat cell type–gene projections forget comparison context. The supplementary note states these results in manuscript form.[89]

B.10 `LLMarkersDB` and marker search

We built `docs/llmarkers.sqlite` from the human-curated benchmark, bioRxiv extractions, and Human Cell Atlas-linked extractions with `analysis/build_llmarkers_sqlite.py`. The database contains paper-level records, marker-level records, optional DEG metrics for benchmark markers, and profile-level records. A profile is one paper, one species, one reported cell type label, its deduplicated marker gene set, and the source sentences supporting those markers.[90]

For text search, each profile has two text fields. The first contains the reported cell type label, paper title, and supporting sentences. The second contains the paper title and abstract. Both fields were embedded offline with `sentence-transformers/all-MiniLM-L6-v2`; normalized float16 vectors were stored in the `profile_embeddings_biomed` table. At query time, the browser embeds the user query with the same MiniLM model through Transformers.js and ranks profiles by a weighted cosine score: 0.9 times similarity to the main profile text plus 0.1 times similarity to the title-and-abstract context. Gene-set search resolves input gene symbols or Ensembl IDs into a query set and ranks profiles by Jaccard similarity between the query set and the stored profile marker set. Both search modes return matched profiles, scores, marker genes, shared genes when applicable, paper metadata, and supporting source text.[91]

B.11 Data and code availability

All code and data are available at `https://github.com/sbooeshaghi/llmarkers`. We also provide an interactive web browser for LLMarkers at `https://sbooeshaghi.github.io/llmarkers/`. The repository includes human-curated marker gene annotations, LLM extractions, differential expression gene lists, and analysis notebooks. The original manuscripts are not redistributed in the repository due to publisher license restrictions; each dataset directory contains a README with the DOI and instructions for obtaining the source paper. The human-curated annotations consist of short factual excerpts (gene names, cell type labels, and brief verbatim rationale sentences) used solely for scientific verification and are intended to comply with fair use.

C Acknowledgements

We thank Ben Johnson, Josh Liang, and Maggie Chasse for a fruitful discussion on cell-types. We also thank Michal Rozenwald Jase Gehring, and Jeremy Marcus, and Kevin An, for helpful feedback on a first draft of these results, as well as folks at the Cancer Genetics and Epigenetics Gordon Research Seminar who gave great feedback on an earlier version of this work. We thank Luca Pinello for suggesting the formaalisation of local vs global marker genes. We also thank all of humanity for their contributions to the training data of the large language models that this work relied on.

D Author contributions

A.S.B. conceived of the study, and compiled the first draft of the benchmark. N.A., along with A.S.B., completed the benchmark and performed initial analyses for verifying correctness of the `LLMarkers`. A.W. suggested expanding the analysis to the bioRxiv and Human Cell Atlas-linked corpora. A.S.B. performed the remaining analysis, along with Claude Opus 4.6, and wrote the manuscript. All authors met and discussed results and reviewed the manuscript.

E Disclosures

All em dashes were placed, in the text, by the author's own hands. Generative language models were used by the authors to check for grammatical errors in the text, fix bugs in code, and debug LaTeX issues.

Claim evidence registry

#	ID	Claim	Sources
1	`abstract-benchmark`	`LLMarkers`, a benchmark of 1,560 reported marker claims from seven single-cell RNA-seq studies, each linked to source evidence and supporting differential expression.	`analysis/benchmark_verification.ipynb`
2	`abstract-deg-recovery`	Reported markers were enriched among strong differential expression signals, but a fixed rank cutoff recovered only about half of them.	`analysis/lfc_comparison.ipynb` `analysis/artifacts/fig2_optimal_deg_cutoff.tsv` `analysis/build_fig2_marker_recovery.py`
3	`abstract-llm-scale`	LLM extraction from manuscripts outperformed zero-shot marker generation and selection from differential expression tables, so we used it to curate 32,621 marker associations from 855 bioRxiv and Human Cell Atlas-linked papers.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py` `analysis/build_marker_corpus.py` `analysis/build_hca_extraction_summary.py`
4	`abstract-cross-study`	Same-named cell types shared marker genes above background, but 99.2% of same-label cross-paper pairs did not report identical marker sets. Recurrent immune labels recovered shared marker vocabularies only in aggregate.	`analysis/artifacts/cross_study_label_marker_joint_distribution.tsv` `analysis/artifacts/fig3_same_label_marker_jaccard_summary.tsv` `analysis/artifacts/fig3_label_local_global_marker_recovery.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
5	`abstract-formal`	We formalize the definition of a marker gene and prove that under this definition it is not self-contained.	`analysis/formal/MarkerIdentifiability/Basic.lean`
6	`abstract-db`	`LLMarkersDB` as a searchable resource that preserves marker claims, mapped genes, source text, and study context.	`docs/llmarkers.sqlite` `analysis/build_llmarkers_sqlite.py` `analysis/export_minilm_embeddings.py` `docs/app.js`
7	`fig-paper-celltype-joint`	*Figure/table source record*	
8	`intro-benchmark-summary`	`LLMarkers`, a benchmark of 1,560 cell type–gene pairs from seven single-cell RNA-seq studies, each linked to the sentence or figure where it was reported and to the differential expression comparison from which it was drawn.	`analysis/benchmark_verification.ipynb`
9	`intro-deg-recovery`	We show that reported markers are enriched among strong differential expression signals, but cannot be recovered by a simple rank threshold.	`analysis/lfc_comparison.ipynb` `analysis/artifacts/fig2_optimal_deg_cutoff.tsv` `analysis/build_fig2_marker_recovery.py`
10	`intro-extraction-best`	Extraction most faithfully recovers the markers that papers report.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
11	`intro-corpus-scale`	Using LLM marker extraction, we curate markers from 504 bioRxiv preprints and 457 Human Cell Atlas-linked publications.	`analysis/build_marker_corpus.py` `analysis/build_hca_extraction_summary.py`

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#	ID	Claim	Sources
12	`intro-corpus-marker-sharing`	The resulting corpus shows that same-named cell types share more marker genes than background, but their marker sets often remain different.	`analysis/artifacts/cross_study_label_marker_joint_distribution.tsv` `analysis/artifacts/fig3_same_label_marker_jaccard_summary.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
13	`intro-db-release`	`LLMarkersDB`, a searchable resource for atlas building that preserves marker claims and source evidence.	`docs/llmarkers.sqlite` `analysis/build_llmarkers_sqlite.py`
14	`benchmark-summary`	Across these studies, `LLMarkers` contains 2,528 markers covering 168 cell types, 641 unique genes, and 1,560 unique cell type–gene pairs.	`analysis/benchmark_verification.ipynb`
15	`benchmark-modality`	1,151 pairs (74%) appeared only in figures, 256 (16%) appeared only in text, and 153 (10%) appeared in both.	`analysis/benchmark_verification.ipynb`
16	`benchmark-deg-overlap`	The overlap between curated markers and DEG tables ranged from 18% in `bone` to 97% in `lung`. In `adipose (Emont)`, the authors describe *PDGFRA* as expressed across all six adipocyte stem and progenitor cell subpopulations, but the DEG table flags *PDGFRA* for only one.	`analysis/benchmark_verification.ipynb`
17	`adipose-name-overlap`	Of the 73 combined cell type names across both studies, only three are shared.	`analysis/benchmark_verification.ipynb`
18	`fig-marker-selection`	*Figure/table source record*	`analysis/build_fig2_marker_recovery.py`
19	`fig-llm-curation`	*Figure/table source record*	`analysis/build_fig2_marker_recovery.py`
20	`benchmark-sign-contradictions`	Of 544 curated marker pairs that appear in at least two DEG tables within the same study, 46 (8.5%) are upregulated in one table and downregulated in another. These contradictions were concentrated in studies with multiple cell-type comparisons or metadata-stratified contrasts, including `adipose (Hildreth)` (39%) and `lung` (12%).	`analysis/benchmark_verification.ipynb`
21	`expert-matched`	Across all seven datasets, 303 text markers and 763 image markers could be matched to DEG entries. All were statistically significant, but their fold changes varied widely (median LFC 1.19, IQR 0.66–2.00).	`analysis/gen_strength.ipynb`
22	`expert-ranks`	The median rank was 40 for text markers and 50 for image markers. About half of reported markers fell within the top 50 DEGs (53% text, 50% image), and about two-thirds fell within the top 100 (66% text, 62% image).	`analysis/lfc_comparison.ipynb`
23	`expert-spearman`	Although text and figure markers were often different genes (median Jaccard 0.33), their per-cell-type fold changes were strongly correlated (Spearman $\rho = 0.80$, $p < 10^{-16}$, $n = 74$), suggesting that marker strength is consistently reported in the data and text.	`analysis/lfc_comparison.ipynb`
24	`hildreth-optimal-n-detail`	The `adipose (Hildreth)` dataset illustrates this pattern: reported markers were strong on average, but the best recovery occurred at $N = 61$ for image markers and $N = 41$ for text markers (**Supplementary Figure S1**).	`analysis/artifacts/fig2_optimal_deg_cutoff.tsv` `analysis/build_fig2_marker_recovery.py`
25	`expert-optimal-n`	Across datasets, the optimal N averaged 76 for image markers and 60 for text markers, with peak F-scores of 0.51 and 0.44, respectively (**Figure 2B**).	`analysis/artifacts/fig2_optimal_deg_cutoff.tsv` `analysis/build_fig2_marker_recovery.py`
26	`gen-pairs`	the model generated 1,390 unique cell type–gene pairs, and 845 (61%) matched at least one DEG entry in the corresponding studies.	`analysis/ext_vs_gen.ipynb` `analysis/gen_strength.ipynb`

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#	ID	Claim	Sources
27	`gen-f1`	Generation achieved a mean pair F1 of 0.17 against expert-curated markers. Performance was lowest for study-specific labels such as "ADIPOCYTE 1" through "ADIPOCYTE 7" in `adipose (Emont)` (F1 = 0.06) and "LBM1" through "LBM3" in `bone` (F1 = 0.19).	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
28	`gen-prediction`	the model predicted the exact reported label for 27 of 181 cell types (15%); in total, 131 of 181 predictions (72%) were exact, substring, or shared-token matches.	`analysis/ext_vs_gen.ipynb`
29	`sel-f1`	but mean pair F1 was 0.16 at each dataset's optimal N, no better than generation.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
30	`sel-bound-eff`	selection reached only 20.2% of the attainable F1 on average.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
31	`ext-results`	we extracted 392 total records, representing 357 within-dataset unique mapped cell type–gene pairs across the seven benchmark studies. Every extracted source sentence matched the manuscript verbatim, and 366 of 392 records (93%) had both the cell type and gene label confirmed within the sentence.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
32	`ext-text-f1`	extraction achieved mean pair F1 of 0.69, with precision 0.71 and recall 0.72.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
33	`ext-data-f1`	mean data-level F1 to 0.61.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
34	`ext-all-f1`	extraction pair F1 fell to 0.32.	`analysis/build_fig2_marker_recovery.py`
35	`method-costs`	It cost \$3.77 total, or \$0.54 per paper, compared with \$2.30 for generation and \$5.94 for selection.	`analysis/selection_analysis.ipynb` `analysis/build_fig2_marker_recovery.py`
36	`fig-cross-study-unification`	*Figure/table source record*	`analysis/build_fig3_local_global_marker_identifiability.py` `analysis/build_cap_liver_local_global_deg.py`
37	`corpus-extraction-summary`	We ran `mrkr` on 504 bioRxiv preprints and 457 Human Cell Atlas-linked publications. Across these corpora, `mrkr` extracted 32,621 marker associations from 855 papers that yielded marker records. The supporting text matched the manuscript exactly, and both the cell type label and gene label were confirmed within the extracted sentence for 89% of bioRxiv records and 93% of HCA-linked records.	`analysis/build_marker_corpus.py` `analysis/build_hca_extraction_summary.py`
38	`corpus-analysis-ready`	For cross-study analysis, we kept verified human markers that mapped to Ensembl gene IDs and deduplicated records within each paper. This produced 24,141 analysis-ready marker records and 3,351 paper-level marker profiles with at least three mapped marker genes, where each profile is a reported cell type label and its mapped marker-gene set in one paper.	`analysis/build_marker_corpus.py` `analysis/cross_study_gene_space.py`
39	`cross-study-same-label-discordance`	Among profile pairs with the exact same normalized label, only 45 had identical marker sets, while 2,408 had partial overlap and 3,167 had no shared marker gene. Thus, 99.2% of same-label pairs did not report the same marker set.	`analysis/artifacts/cross_study_label_marker_joint_distribution.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`

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#	ID	Claim	Sources
40	`cross-study-different-label-overlap`	Across profiles with different labels, 102,513 pairs shared at least one marker gene.	`analysis/artifacts/cross_study_label_marker_joint_distribution.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
41	`immune-same-label-discordance`	Among cross-paper profile pairs with the same label, 54% of "T cell" pairs, 64% of "CD8 T cell" pairs, 65% of "CD4 T cell" pairs, 70% of "macrophage" pairs, and 74% of "monocyte" pairs shared no marker gene.	`analysis/artifacts/fig3_same_label_weak_marker_examples.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
42	`different-label-shared-marker-examples`	"B cell" and "B cells" shared *CD79A*, *MS4A1*, and *CD19*; "monocyte" and "monocytes" shared *S100A8*, *S100A9*, and *CD14*; and "naive T cell" and "TCM" shared *TCF7*, *CCR7*, *LEF1*, and *SELL*.	`analysis/artifacts/fig3_different_label_shared_marker_examples.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
43	`cap-cross-project-same-label-discordance`	Among the HCA's Cell Annotation Platform (CAP) human profile pairs from different projects with the exact same reported label, only 2 of 307 pairs (0.7%) had identical marker sets, while 97 pairs (31.6%) shared no marker gene.	`analysis/artifacts/fig3_cap_cross_project_label_marker_joint_distribution.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
44	`cap-label-same-label-marker-sharing`	Across 307 cross-project same-label CAP profile pairs, 68% shared at least one marker gene and the mean marker-set Jaccard was 0.161, compared with 2% and 0.003 for random different-label pairs.	`analysis/artifacts/fig3_same_label_marker_jaccard_summary.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
45	`immune-local-global-recovery`	"T cell" had mean local specificity 0.90 and mean global recovery 0.80, "B cell" had 0.91 and 0.70, "TREG" had 0.73 and 0.69, "macrophage" had 0.93 and 0.60, and "monocyte" had 0.88 and 0.46.	`analysis/artifacts/fig3_label_local_global_marker_recovery.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
46	`llmarkers-same-label-marker-sharing`	Across 5,620 cross-paper same-label LLMarkers profile pairs, 44% shared at least one marker gene and the mean marker-set Jaccard was 0.098, compared with 2% and 0.003 for random different-label pairs. The median same-label Jaccard was still 0.	`analysis/artifacts/fig3_same_label_marker_jaccard_summary.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
47	`cap-ontology-local-global-recovery`	Across 78 recurrent CAP ontology terms, median local specificity was 0.72 and median global recovery was 0.33; "B cell" had mean local specificity 0.92 and mean global recovery 0.43, while "neutrophil" had 0.89 and 0.79.	`analysis/artifacts/fig3_cap_ontology_local_global_marker_recovery.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
48	`cap-liver-local-global-de`	Across 10 local myeloid labels, local DE recovered 74% of reported markers in the top 100 on average, while global DE recovered 21%; every local label had higher local than global recovery. The mean maximum F1 for reported marker retrieval was 0.24 locally and 0.07 globally.	`analysis/artifacts/cap_liver_local_global_deg_summary.tsv` `analysis/artifacts/cap_liver_local_global_recovery_subsampling_summary.tsv` `analysis/build_cap_liver_local_global_deg.py`
49	`formal-marker-definition`	The formalization shows that cell-type marker sets are intersections of pairwise marker sets, and that broadening the comparison family can only remove markers.	`analysis/formal/MarkerIdentifiability/Basic.lean`
50	`profile-search-count`	The database contains 10,132 paper-level marker profiles, including 183 profiles from the human-curated benchmark, 3,805 profiles from the bioRxiv corpus, and 6,144 profiles from Human Cell Atlas-linked publications.	`docs/llmarkers.sqlite` `analysis/build_llmarkers_sqlite.py`
51	`profile-search-results`	`LLMarkersDB` supports free-text queries, such as "cytotoxic lymphocytes" or "stromal fibroblasts," using text embeddings, and gene-set queries using Jaccard similarity over marker genes.	`analysis/export_minilm_embeddings.py` `docs/app.js`
52	`tab-benchmark`	*Figure/table source record*	`analysis/benchmark_verification.ipynb`

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#	ID	Claim	Sources
53	`fig-hildreth-detail`	*Figure/table source record*	`analysis/lfc_comparison.ipynb`
54	`tab-llm-curation`	*Figure/table source record*	`analysis/ext_vs_gen.ipynb` `analysis/selection_analysis.ipynb` `analysis/llm_extraction_eval.ipynb` `analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
55	`tab-monocyte`	*Figure/table source record*	`analysis/monocyte_table_provenance.ipynb` `analysis/artifacts/tables/monocyte_different_marker_genes_provenance.csv`
56	`tab-similar-marker-profiles`	*Figure/table source record*	`analysis/cross_study.ipynb` `analysis/artifacts/cross_study_row3_name_judgment_diff_labels_j1.tsv` `analysis/artifacts/cross_study_context_comparison_review_anchored.md`
57	`methods-benchmark-schema`	The LLMarkers benchmark was assembled by manual curation of seven single-cell RNA-seq studies spanning six tissues: `adipose` [Emont et al., 2022, Hildreth et al., 2021], `bone` [Zhong et al., 2020], `eye` [Gautam et al., 2021], `lung` [Adams et al., 2020], `ovary` [Wagner et al., 2020], and `testis` [Shami et al., 2020]. The benchmark was designed as a carefully linked evaluation set rather than a comprehensive survey of single-cell reporting styles. For each study, a curator read the manuscript and recorded every marker gene–cell type association, the verbatim sentence or figure panel from which it was drawn (source_rationale), whether it came from text or a figure (source_type), the supplementary data source containing the corresponding differential expression results (data_id), and the gene's Ensembl identifier (feature_id). Gene symbols were mapped to Ensembl IDs using `mrkr map`, which queries the Ensembl REST API.	`analysis/benchmark_verification.ipynb`
58	`methods-deg-parsing`	Each study's supplementary tables were parsed to extract differential expression gene lists. Header detection was performed automatically by scanning the first 10 rows for a row containing expected column names (e.g., gene, p_value, avg_log2FC). The resulting DEG lists were stored in the same evidence schema as human annotations, enabling direct comparison.	`analysis/benchmark_verification.ipynb`
59	`methods-verification`	Across all seven datasets, 96.2% of source rationales matched the manuscript exactly. We additionally verified that the group label (cell type name) and feature label (gene symbol) could be localized within each source rationale using the `taln` text alignment library, which performs subword-level alignment. Group labels were confirmed in 97.5% of records and feature labels in 99.5%.	`analysis/benchmark_verification.ipynb`

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#	ID	Claim	Sources
60	`methods-marker-strength-matching`	To quantify where human-curated marker genes fall in the differential expression ranking, we matched each marker to its corresponding entry in the dataset's DEG list. For human-curated markers, matching was performed on the four-part key (dataset, data_id, group_name, feature_id), requiring exact agreement on the cell type, gene, and supplementary data source. For LLM-generated markers, which have no data_id, we matched on (group_name, feature_id) against all DEG data sources within each dataset. Markers that could not be matched—because the gene was absent from the DEG list or lacked an Ensembl ID—were excluded from this analysis.	`analysis/gen_strength.ipynb`
61	`methods-marker-rank-summary`	For each matched marker, we recorded its rank within the cell type's DEG list (rank 1 = highest log-fold change) and its log-fold change. We computed the fraction of markers falling within the top N DEGs for thresholds $N \in \{10, 25, 50, 100\}$ and separately tabulated these fractions for text-sourced markers, image-sourced markers, and generated markers.	`analysis/lfc_comparison.ipynb`
62	`methods-marker-permutation`	To test whether discussed markers are drawn from stronger DEGs than expected by chance, we performed permutation tests at the level of individual cell types within each dataset and data source. For each cell type with at least two discussed markers, we sampled a size-matched set of genes from the top 100 significant ($p < 0.05$) DEGs that were not in the human-curated marker set. We then compared the mean rank of the discussed markers against the null distribution obtained by randomly partitioning the combined set 10,000 times. Tests were two-sided at $\alpha = 0.05$.	`analysis/lfc_comparison.ipynb`
63	`methods-optimal-deg-depth`	To estimate the optimal DEG list depth for recovering each marker set, we computed precision–recall curves by varying a rank threshold N from 1 to 20,000. At each threshold, we defined the top N DEGs as predicted markers and computed precision, recall, and the F-score. The optimal N was defined as the threshold that maximized the F-score. This analysis was performed separately for each dataset and marker source (text, image, or generated).	`analysis/artifacts/fig2_optimal_deg_cutoff.tsv` `analysis/build_fig2_marker_recovery.py`
64	`methods-marker-source-tests`	Differences in rank and log-fold change between marker sources were assessed using two-sided Mann–Whitney U tests.	`analysis/lfc_comparison.ipynb`
65	`methods-llm-run-settings`	We evaluated three approaches to LLM-based marker gene curation across the seven benchmark datasets. All experiments used Claude (claude-opus-4-6) with temperature 0 and a maximum output length of 32,000 tokens; we did not benchmark model-to-model variation. Costs were computed from the model's API pricing ($5 per million input tokens, $25 per million output tokens for claude-opus-4-6).	`analysis/build_fig2_marker_recovery.py` `analysis/selection_analysis.ipynb`
66	`methods-llm-generation`	In the generation experiment (celltypes-to-genes), we gave the model the list of cell type names from each dataset's human-curated marker set and asked it to generate 5–15 well-established marker genes per cell type. The model received only the species and cell type names—no manuscript text, figures, or supplementary data. In a second generation experiment (genes-to-celltypes), we extracted the gene list for each cell type from the human-curated set, replaced the cell type names with anonymous labels (Group 1, Group 2, ...), and asked the model to predict the cell type for each group. We measured prediction accuracy as the fraction of exact string matches between predicted and original cell type names. Non-exact predictions were counted as related when they showed substring containment or shared label tokens; remaining predictions were counted as having no lexical relation.	`analysis/ext_vs_gen.ipynb`

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#	ID	Claim	Sources
67	`methods-llm-selection`	In the selection experiment, we parsed each dataset's DEG tables, ranked genes per cell type by absolute log fold change, and presented the top N genes (with log fold change and adjusted p-value) to the model, asking it to select which are true markers. We swept $N \in \{10, 20, 30, 40, 50, 100, 200, 300, 400, 500\}$ for each dataset and reported the N that maximized pair-level F1. As a control for whether cell type identity aids selection, we repeated the full sweep with cell type names replaced by anonymous cluster labels (Cluster 1, Cluster 2, . . .), preserving the gene lists and statistics but removing biological context from the names.	`analysis/selection_analysis.ipynb` `analysis/build_fig2_marker_recovery.py`
68	`methods-llm-extraction`	In the extraction experiment, we ran the `mrkr extract` pipeline on each manuscript, followed by `mrkr verify` and `mrkr map`. We compared the extracted evidence against human curation at three levels of stringency. At the feature level, we computed the Jaccard similarity between unique gene sets (Ensembl gene IDs), ignoring cell type assignment. At the pair level, we defined a match as a (group_name, feature_id) pair present in both sets and computed precision, recall, and F1. At the data level, we required a full (data_id, group_name, feature_id) triple to match. We also computed a mean per-cell-type gene F1 by averaging the gene-level F1 across all cell types shared between the LLM and human sets.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
69	`methods-llm-evaluation-subset`	Because the LLM extracts only from manuscript text while human curators also annotated markers from figures and supplementary tables, we evaluated all three methods—generation, selection, and extraction—against the subset of human annotations with source_type = "text" (**Supplementary Table S2**; **Figure 2**). We additionally report extraction metrics against all annotations to quantify the effect of image-only markers. All comparisons used Ensembl gene IDs (feature_id) rather than gene symbols to avoid ambiguity from gene name variants.	`analysis/artifacts/fig2_llm_recovery.tsv` `analysis/build_fig2_marker_recovery.py`
70	`methods-llm-source-verification`	Source verification statistics were derived from the `_verification` field attached to each LLM extraction by `mrkr verify`. This field records whether the source rationale was found verbatim in the manuscript text, and whether the group label and feature label could be aligned within the source rationale using the `taln` text alignment library.	`analysis/llm_extraction_eval.ipynb` `analysis/build_fig2_marker_recovery.py`
71	`methods-biorxiv-mirror`	We selected bioRxiv preprints for marker gene extraction from a local mirror of 428,741 MECA archives downloaded from the bioRxiv source repository (s3://biorxiv-src-monthly/, accessed January 2025). Each MECA archive contains the manuscript's JATS XML, from which we extracted structured metadata including title, abstract, author-provided keywords, DOI, and license.	`analysis/biorxiv_summary.ipynb` `analysis/build_marker_corpus.py`
72	`methods-biorxiv-filter`	We filtered manuscripts using three criteria. First, we required a permissive license (CC-BY or CC0) to allow downstream text processing and redistribution. Second, we required evidence of human relevance by matching the abstract or author-provided keywords against terms including "human", "Homo sapiens", "patient", "clinical", and "PBMC". Third, we required relevance to cell-type marker genes by matching against two sources: (i) author-provided JATS keywords, available for 49% of manuscripts, matching terms such as "marker gene", "cell type", "scRNA", "single-cell RNA", "deconvolution", and "cell atlas"; and (ii) abstract text, matching phrases such as "marker gene", "cell-type marker", "cell type annotation", and co-occurrences of "single-cell" or "scRNA-seq" with "marker." Manuscripts were included if they matched on either keywords or abstract text. When multiple versions of the same manuscript existed (identified by shared DOI), we retained only the most recent version. This procedure yielded 504 unique manuscripts.	`analysis/biorxiv_summary.ipynb` `analysis/build_marker_corpus.py`

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73	`methods-biorxiv-extraction-workflow`	Each manuscript's JATS XML was converted to markdown using a custom JATS parser that preserves article structure (sections, headings, inline figure captions) while stripping references and HTML markup. We then ran `mrkr extract` on each markdown file using text-only extraction (no figures), followed by `mrkr verify` and `mrkr map`. Papers were processed in parallel (25 concurrent jobs). Papers yielding zero extractions were recorded with empty marker files.	`data/biorxiv/convert_all.sh` `data/biorxiv/run_mrkr_all.sh` `analysis/build_marker_corpus.py`
74	`methods-hca-publication-corpus`	We compiled a second corpus from publications linked to Human Cell Atlas projects. HCA project metadata were used to collect publication titles, URLs, and DOIs. Records were deduplicated by DOI when available and otherwise by normalized title. This produced 515 HCA-linked publication records, of which 475 had cached markdown manuscript text under `data/hca/manuscripts`. The cached files were normalized to the same plain-text format used for the bioRxiv corpus.	`analysis/compile_hca_human_publications.py` `analysis/download_hca_manuscripts.py` `analysis/unify_hca_manuscripts.py` `data/hca/manuscripts_manifest.tsv`
75	`methods-hca-extraction-workflow`	We processed the cached HCA-linked manuscripts with the same extraction workflow used for bioRxiv: `mrkr extract`, followed by `mrkr verify` and `mrkr map`. The run produced marker-output files for 457 publications, of which 434 contained at least one marker record. Papers yielding no marker records were kept in corpus summaries but did not contribute marker profiles. Cross-study analyses used only records that were human, verified against source text, mapped to Ensembl gene IDs, and retained after within-paper deduplication.	`analysis/build_hca_extraction_summary.py` `data/hca/run_mrkr_all.sh` `data/hca/manuscripts_manifest.tsv`
76	`methods-gene-id-mapping`	Gene symbols were mapped to Ensembl identifiers using a reference file built from Ensembl GRCh38.114. Canonical gene names were extracted from the Ensembl GTF annotation, and gene synonyms were retrieved from the Ensembl REST API, following the approach used by gget [Luebbert and Pachter, 2023]. The resulting mapping file contains 94,000 unique gene names and synonyms mapping to 40,000 unique Ensembl gene IDs. Lookup is case-insensitive. Only human records (organism = "homo_sapiens") were mapped; non-human records were left without Ensembl identifiers.	`analysis/cross_study_gene_space.py` `analysis/build_marker_corpus.py`
77	`methods-cap-profile-construction`	We constructed an independent set of marker profiles from Cell Annotation Platform marker records in `data/cell_annotation_platform/markers.json`. CAP records were restricted to human records with Ensembl gene IDs, standardized with the same gene-mapping utilities used for `LLMarkers`, and grouped by CAP dataset, project, label set, and group name. A CAP marker profile was retained when the group had at least three mapped marker genes. We stored the author-provided group label, CAP dataset and project identifiers, label-set name, source ID, Cell Ontology term and category annotations, and rationale DOI metadata when present.	`data/cell_annotation_platform/markers.json` `data/cell_annotation_platform/cap_download_manifest.tsv` `analysis/build_cap_llmarkers_comparison.py` `analysis/artifacts/cap_llmarkers_resource_summary.tsv`
78	`methods-cap-ontology-profiles`	For ontology-based CAP analyses, we restricted CAP profiles to records with a non-empty Cell Ontology term and a CL identifier, then used the ontology term rather than the author-reported label for label normalization and cross-project grouping.	`analysis/build_fig3_local_global_marker_identifiability.py` `analysis/artifacts/fig3_cap_ontology_local_global_marker_recovery.tsv`
79	`methods-cross-paper-profile-construction`	To compare local and global marker definitions, we constructed one binary marker vector for each (paper, reported cell type label) pair. The vector was the set of mapped Ensembl gene IDs reported for that label in that paper. Profiles with fewer than three mapped marker genes were excluded from cross-paper analyses. All analyses were restricted to records that were human, verified against the source text, mapped to Ensembl gene IDs, and deduplicated within each paper.	`analysis/cross_study_gene_space.py` `analysis/build_fig3_local_global_marker_identifiability.py`

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80	`methods-label-normalization`	Cell type labels were normalized by converting to ASCII, uppercasing, replacing non-alphanumeric characters with spaces, splitting letter-number boundaries, and standardizing spacing. Exact label pairs had identical normalized labels. Partial label pairs had one normalized label contained in the other, or shared at least one non-generic token after removing generic terms such as "cell", "cells", "type", and "cluster". All remaining label pairs were labeled different. This rule is string-based and does not use ontology mapping or semantic similarity.	`analysis/build_fig3_local_global_marker_identifiability.py`
81	`methods-marker-similarity-bins`	Marker-gene set similarity was measured with Jaccard similarity, defined as the number of shared Ensembl gene IDs divided by the size of the union of the two marker sets. Marker pairs were assigned to three bins: exact if Jaccard similarity was 1, partial if Jaccard similarity was greater than 0 and less than 1, and none if Jaccard similarity was 0. The joint distributions in **Figure 3A,D** count unordered cross-paper or cross-project profile pairs across the label-relation and marker-relation bins.	`analysis/artifacts/cross_study_label_marker_joint_distribution.tsv` `analysis/artifacts/fig3_cap_cross_project_label_marker_joint_distribution.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
82	`methods-same-label-marker-sharing`	For the same-label marker-sharing analysis in **Figure 3B,E**, we computed all cross-paper or cross-project pairwise Jaccard similarities among profiles with the same normalized label. As a background comparison, we sampled 100,000 cross-paper or cross-project profile pairs with different normalized labels and computed the same Jaccard statistic. Plotted points were sampled for display, while means and percentages were computed from the full pair set.	`analysis/artifacts/fig3_same_label_marker_jaccard_summary.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
83	`methods-local-global-recovery`	For the local/global recovery analysis in **Figure 3C,F**, we first identified, for each profile, the markers that were specific within the source paper or CAP project. A marker was locally specific if it appeared in that profile and was not reported for any other label in the same paper or project. For each recurrent label or ontology term, we then asked whether those locally specific markers appeared in any same-label or same-term profile from another paper or project. Local specificity is the mean fraction of profile markers that are locally specific. Global recovery is the mean fraction of locally specific markers that are recovered in at least one same-label or same-term profile from another paper or project. LLMarkers labels were included when they appeared in at least five profiles. CAP ontology terms were included when they appeared in at least three profiles.	`analysis/artifacts/fig3_label_local_global_marker_recovery.tsv` `analysis/artifacts/fig3_cap_ontology_local_global_marker_recovery.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`
84	`methods-different-label-examples`	For the different-label examples in **Figure 3B,E**, we searched cross-paper or cross-project pairs with different normalized reported labels and at least three shared marker genes. We grouped candidate pairs by label pair and retained examples with repeated evidence across at least two profile pairs.	`analysis/artifacts/fig3_different_label_shared_marker_examples.tsv` `analysis/build_fig3_local_global_marker_identifiability.py`

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#	ID	Claim	Sources
85	`methods-cap-liver-expression-analysis`	For the CAP liver expression-data analysis in **Figure 3G–I**, we used CAP dataset 1440 as the local liver myeloid context and CAP dataset 1437 as the broader all-cells liver context. We loaded each AnnData file with `anndata.read_h5ad`, used the `user_provided` expression layer when present, and used `author_cell_type` as the cell type label. For each local myeloid label, we computed one-vs-rest DE scores for all genes as the difference in mean expression divided by its standard error. We then mapped local labels to their broader all-cells labels using the CAP author labels and compared the local one-vs-rest ranking to the corresponding global one-vs-rest ranking. Reported markers were taken from the CAP marker JSON, restricted to human Ensembl-mapped genes. Top-100 recovery is the fraction of reported markers appearing in the top 100 genes by DE score. For **Figure 3G**, we recomputed the DE rankings across 50 cell-type-stratified subsampling replicates and plotted 2.5th–97.5th percentile intervals. Each replicate sampled 80% of cells within each `author_cell_type`, capped at 10,000 cells per cell type. DE stability is the Jaccard similarity between the top 100 local DE genes and the top 100 global DE genes. Marker retrieval F1 was computed over the ranked DE list by treating reported markers as positives and reporting the maximum F1 over rank cutoffs.	`analysis/artifacts/cap_liver_local_global_deg_summary.tsv` `analysis/artifacts/cap_liver_local_global_recovery_subsampling_summary.tsv` `analysis/build_cap_liver_local_global_deg.py`
86	`methods-figure-one-examples`	The examples in **Figure 1** were selected from the extracted corpus to illustrate the label and marker relations used in the cross-paper analysis. The toy matrices are schematic. The displayed myeloid marker profiles are real extracted profiles, and their joint distribution was computed only over the displayed profiles using the same label-relation and marker-relation rules described above.	`docs/paper/biorxiv/tex-figures/fig_paper_celltype_joint.tex` `docs/paper/biorxiv/tex-figures/fig_paper_celltype_joint_body.tex`
87	`methods-supplementary-profile-tables`	For **Supplementary Table S3**, we selected bioRxiv profiles whose normalized reported label contained “monocyte”, retained profiles with at least three mapped marker genes, and kept one profile per paper by choosing the monocyte-labeled profile with the most extracted markers. For **Supplementary Table S4**, we searched cross-paper profile pairs with different reported labels and identical mapped marker sets. We then selected representative exact marker-profile neighborhoods and report the source paper, reported label, source context, and shared marker set. The table reports observed label–marker separation patterns rather than biological interpretations or formal ontology mappings.	`analysis/monocyte_table_provenance.ipynb` `analysis/artifacts/tables/monocyte_different_marker_genes_provenance.csv` `analysis/cross_study.ipynb` `analysis/artifacts/cross_study_row3_name_judgment_diff_labels_j1.tsv` `analysis/artifacts/cross_study_context_comparison_review_anchored.md`
88	`methods-formalization-setup`	The formal results were encoded in Lean 4 under `analysis/formal`. The basic model treats reported marker evidence as a binary marker matrix over finite cell-type, profile, or context-conditioned groups and marker genes. A marker panel separates a comparison family when every distinct pair of groups has different binary signatures on at least one selected gene. The contrast-level model additionally records study, partition, target group, comparison family, gene, threshold, pairwise contrasts, background contrasts, and real-valued marker scores. We used `span` to align supplementary-note statements with Lean declarations and inspect the paper-facing dependency graph [Pachter, 2026].	`analysis/formal/MarkerIdentifiability/Basic.lean` `analysis/formal/CLAIMS.md` `analysis/formal/lakefile.lean` `analysis/formal/lean-toolchain` `docs/paper/biorxiv/build/supplementary-note.lea`
89	`methods-formalization-theorems`	The Lean development proves the marker-family intersection theorem, monotonicity of marker sets under comparison-family restriction, local/global and hierarchy inclusion results, one-vs-all equivalence at the binary pairwise level, weighted-background contrast identities, score-thresholded marker-family results, and examples showing that flat cell type–gene projections forget comparison context. The supplementary note states these results in manuscript form.	`analysis/formal/MarkerIdentifiability/Basic.lean` `analysis/formal/CLAIMS.md` `docs/paper/biorxiv/supplementary-note.tex`

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90	`methods-db-build`	We built `docs/llmarkers.sqlite` from the human-curated benchmark, bioRxiv extractions, and Human Cell Atlas-linked extractions with `analysis/build_llmarkers_sqlite.py`. The database contains paper-level records, marker-level records, optional DEG metrics for benchmark markers, and profile-level records. A profile is one paper, one species, one reported cell type label, its deduplicated marker gene set, and the source sentences supporting those markers.	`docs/llmarkers.sqlite` `analysis/build_llmarkers_sqlite.py`
91	`methods-db-search`	For text search, each profile has two text fields. The first contains the reported cell type label, paper title, and supporting sentences. The second contains the paper title and abstract. Both fields were embedded offline with `sentence-transformers/all-MiniLM-L6-v2`; normalized float16 vectors were stored in the `profile_embeddings_biomed` table. At query time, the browser embeds the user query with the same MiniLM model through Transformers.js and ranks profiles by a weighted cosine score: 0.9 times similarity to the main profile text plus 0.1 times similarity to the title-and-abstract context. Gene-set search resolves input gene symbols or Ensembl IDs into a query set and ranks profiles by Jaccard similarity between the query set and the stored profile marker set. Both search modes return matched profiles, scores, marker genes, shared genes when applicable, paper metadata, and supporting source text.	`analysis/export_minilm_embeddings.py` `docs/app.js`

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